

Institutions versus Markets in Information Aggregation and the Theory of the Firm*

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Abstract

Studies on rational expectation teach us that the market could be viewed as an information aggregation device through equilibrium price. However, because of the noise embedded in a large market, institutional innovations necessarily arise. In particular, I show that market participants always have incentives to create partnerships among themselves. Optimal profit-sharing within such a partnership replicates the effect of direct communication, but it is immune to falsification or high communication cost. This insight renders a new theory of the firm based on *de facto* knowledge aggregation, which is particularly useful for understanding the increasingly dominant human capital intensive firms in a knowledge economy. I discuss several implications of this new perspective on general corporate finance topics.

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Among the many roles a well-functioning capital market plays in an economy, the aggregation of (dispersed) information among market participants has long been emphasized. Classic papers of noisy rational expectation equilibria (e.g. [Grossman and Stiglitz \(1980\)](#), [Hellwig \(1980\)](#), [Diamond and Verrecchia \(1981\)](#)) analyzes how market participants make investment decision based on not only their private information, but also inferences of others' information gleaned from market prices. This information aggregation function of the market is important, because investors (entrepreneurs) usually only have imprecise knowledge of the value of the assets (projects) they seek to hold (invest in). If their own assessments were all they had to rely on, risk would deter them from holding (investing in) all but the most trivial amount. Fortunately, because rational expectation inference reduces the conditional variance of investment payoffs, larger positions are taken, costs of capital are lowered, and entrepreneurship is encouraged. In this sense, a capital market could be viewed as a substitute for information sharing.

Despite its elegance, the noise embedded in a large market guarantees that rational expectation inference is by nature imperfect.¹ Such imperfectness makes noisy rational expectation equilibrium a somewhat constrained outcome, and suggests that non-market information aggregation mechanisms will necessarily coexist with a market economy. For example, direct communication among informed investors will sharpen their information precision and improve information aggregation solely provided by the market.²

However, in reality there are at least two obstacles to direct communication. First, truthful communication may not be incentive compatible. The fear of one's valuable information being abused against, as well as the jeopardy of unintentional divulgence or blatant re-sale

¹See [Black \(1986\)](#) for a comprehensive assessment of "noise". Existing literature often attributes the non-informative "noise" to quantity shocks, yet the noise could be alternatively interpreted as a reduced-form description of investors' imperfect inference from price due to their incomplete knowledge of the large, anonymous market that they are part of.

²This insight is seen in the "information percolation" literature, which explicitly models the evolution of investors' information precision over repeated direct communication, see [Duffie, Giroux, and Manso \(2010\)](#) and [Andrei and Cujean \(2013\)](#), etc.

by other informed parties, often deter or distort truthful communication incentives.³ Second, communication takes time. If a market opportunity is short-lived and requires immediate reaction, delays caused by direct communication might negate its benefit.

My paper points out that an alternative mechanism, which resembles the formation of a partnership firm, could replicate the information sharing effect of direct communication without actually incurring it, thus resolving the disincentive against truthful communication. In this mechanism, investors partner together and write *ex ante* contracts to share total investment payoff that will be realized *ex post*. Because the partnership contract alters each partners' incentive, it changes their investment behavior as well as final payoff. As will be shown, when the *ex ante* sharing rule is optimally designed, each partners' payoff would be exactly identical to that under (hypothetically) truthful, instant, and costless direct communication. A partnership facilitates *de facto* communication among its partner when actual information sharing is costly, if not infeasible.⁴

The fact that information aggregation can be achieved without actual communication might look surprising at first sight, but the intuition behind is indeed straightforwardly rooted in the conventional wisdom that "actions speak louder than words".⁵ Even if investors cannot *ex ante* communicate, their information is nevertheless reflected in their actions, and further reflected in their *ex post* payoffs. Via properly mixing investors' payoffs, in equilibrium their

³Such friction is brought up in seminal discussions on information production (e.g. [Hirshleifer \(1971\)](#)), and motivates studies on optimal selling of information (e.g. [Admati and Pfleiderer \(1986\)](#) and [Admati and Pfleiderer \(1990\)](#)). [Allen \(1990\)](#) develops a theory of financial intermediation based on indirect selling of information. See [Dewatripont and Tirole \(2005\)](#) for further discussions on potential costs and biases in knowledge transfers.

⁴This *de facto* information aggregation effect shall be distinguished from truthful *actual* communication incentives also recovered by partnering. As a sharing contract links each partner's compensation to the payoff of the entire partnership, or the joint investment decisions of all partners, each partner internalizes the benefit her truthful communication on others. This intuition underlies e.g. [Garicano and Santos \(2004\)](#) on how partnership encourages efficient case referral among lawyers. Note however, this channel only resolve the truth-telling problem but not the cost or delay involved in *actual* communication, while the *de facto* information communication channel achieves them all.

⁵The intuition that actions may credibly convey information is also seen in other economic setting, for example, signaling games (e.g. [Leland and Pyle \(1977\)](#)) and the revelation principle in mechanism design.

private information gets indirectly aggregated.

That a partnership complements the market in aggregating information sheds new light on an old question in the theory of the firm: what is the nature of the firm, and what additional role a firm provides to pure market allocation? As a joint-stock firm could be viewed as a payoff-sharing partnership among its various stakeholders, one of the reasons that a firm exist could very well be to offer a non-price mechanism for facilitating knowledge aggregation among its stakeholders. In another word, firms and markets coexist because of their complementarity in implementing the wisdom of crowds.

This perspective is particularly relevant in today’s knowledge economy. Although the wisdom of crowds has always been guiding entrepreneurial endeavors, never has it been more important than it is today. In face of a knowledge economy inundated by a flood of information, a single person’s knowledge often falls short of what is required to secure adequate confidence of business success. The extinction of polymaths calls for more attention to studies on knowledge aggregation. Interpreting firms as *de facto* knowledge aggregators, my result help clarify how efficient knowledge aggregation takes place within and across firms, as well as instruct us how to better design organizational structures to secure knowledge aggregation. The results have further implications for general corporate finance topics on corporate governance, capital structure, and valuation. In particular, the compensation and governance structure of the partnerships described in my model might shed light on recent advocates for “workplace democracy” as well as employee ownership.

The rest of the paper is organized as follows. Section 1 derives the endogenous arise of a partnership in the market under a noisy rational expectation setup, based on which Section 2 proposes a new theory of the firm showing on how firm creation promotes stakeholders dedication to their work (compared to running sole-proprietors). Section 3 discusses general implications of the sharing partnership theory of the firm on various corporate governance topics. Section 4 reviews and relates existing literature. Section 5 concludes.

1 Partnership and *de facto* Information Aggregation

This section describes a partnership firm in simplified noisy rational expectation model setup à la Hellwig (1980). I show that a partnership achieves *de facto* information aggregation without actual communication. While the market also communicates information through equilibrium price, it is nevertheless dominated by the *de facto* communication within a partnership.

Similar to Hellwig (1980), the market consists of a continuum of entrepreneurs with constant absolute risk aversion (CARA) utility of risk aversion ρ and wealth w . From time to time, new investment opportunities arrive. For simplicity, consider a two-period model with no discounting. On $t = 0$, a risky investment project in need of minimum financing of I arises. The entrepreneurs decide on x , the optimal number of shares to invest in the project. The investment project has *ex ante* valuation of $v \sim \mathcal{N}(\bar{v}, \tau_v^{-1})$ that is determined by nature on $t = 0$ but is not known to anyone until $t = 1$. When making her investment decision, each entrepreneur j is endowed with a signal of the project valuation $i_j = v + e_j$, where v and e_j are independent and $e_j \sim \mathcal{N}(0, \tau_e^{-1})$. The private signal is interpreted as the entrepreneur's private information or expertise/knowledge (in assessing the prospect of the new investment opportunity). All other uncertainties regarding the structure of the market is summarized in a quantity noise z , which appears like *per-capita* random demand of the project where $z \sim \mathcal{N}(\bar{z}, \sigma_z^2)$ and carries no information about the project valuation (i.e., $z \perp v$).

A partnership of size n is defined by a charter that specifies *ex ante* that n entrepreneurs equally receive $\frac{1}{n}$ of any *ex post* realized investment payoff. Because entrepreneurs are symmetric in terms of their signal precision, the payoff-sharing agreement is an *ex ante* fair deal. The sharing feature captures the reality that each entrepreneur's compensation is based on proportional ownership (in this case equal ownership) in the firm, which is often determined before new investment opportunity emerges.

The $\frac{1}{n}$ revenue-sharing rule is a natural outcome of the symmetric assumption imposed for ease of exposition. Appendix D solves a general version when both entrepreneurs' informedness and risk preference are heterogeneous. In such asymmetric cases, as long as the sharing rule is optimally designed (which will be an asymmetric sharing rule based solely on partner's risk preference), the *de facto* information aggregation effect will be obtained without change.

1.1 Investment Behaviors within a Partnership

I first present a general result on the optimal investment demand and expected utility for an entrepreneur in a size- n firm given an arbitrary linear valuation (price) system.

Theorem 1.1. *In an economy in which the equilibrium project valuation follows a linear function $p = \mu + \pi v - \gamma z$, the optimal investment demand of an entrepreneur j in a firm of size n is given by*

$$x_j = \frac{1}{\rho} \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n \tau_e i_j - (n \tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right] \quad (1)$$

and her expected utility

$$- \frac{\exp(\frac{A}{2B})}{\sqrt{B}}, \text{ where} \quad (2)$$

$$\begin{aligned} A &= - \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + \tau_e i_j - (\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right]^2 \frac{n \tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{(\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v)^2} \\ B &= \frac{n \tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v} \end{aligned}$$

Proof. See Appendix. □

1.2 The Incentive of Partnership Creation in a Market Economy

In this section, I apply Theorem 1.1 to a Hellwig (1980) market to show that partnership creation is an endogenous outcome in a market outcome, i.e. with rational entrepreneurs a market economy does not exist in isolation.

In a Hellwig (1980) market, the equilibrium price is given by (see e.g. Ross (2005))

$$p = \mu + \pi v - \gamma z \quad (3)$$

where

$$\begin{aligned} \mu &= \frac{\bar{v}\sigma_z^2 - \tau_v^{-1}\bar{z}\frac{\tau_e}{\rho}}{\sigma_z^2 + \tau_v^{-1}\tau_e\sigma_z^2 + \tau_v^{-1}\left(\frac{\tau_e}{\rho}\right)^2} \\ \pi &= \frac{\tau_v^{-1}\tau_e}{\sigma_z^2 + \tau_v^{-1}\tau_e\sigma_z^2 + \tau_v^{-1}\left(\frac{\tau_e}{\rho}\right)^2} \left[\sigma_z^2 + \frac{\tau_e}{\rho^2}\right] \\ \gamma &= -\frac{\rho\tau_v^{-1}}{\sigma_z^2 + \tau_v^{-1}\tau_e\sigma_z^2 + \tau_v^{-1}\left(\frac{\tau_e}{\rho}\right)^2} \left[\sigma_z^2 + \frac{\tau_e}{\rho^2}\right] \end{aligned}$$

Plug in Theorem 1.1, we have,

Corollary 1.2. *In a Hellwig (1980) market economy, if n entrepreneurs among the entrepreneur continuum meet and create a partnership firm, a particular partner j 's equilibrium investment is given by*

$$x_j = \frac{1}{\rho} \left[\rho \frac{\rho\sigma_z^2\tau_v\bar{v} - \tau_e\bar{z}}{\rho^2\sigma_z^2 + \tau_e} + n\tau_e i_j - \left(n\tau_e + \frac{\rho^2\sigma_z^2\tau_v}{\rho^2\sigma_z^2 + \tau_e} \right) p \right]. \quad (4)$$

Comment: If entrepreneur j instead choose to keep running as a sole-proprietor, her investment would be given by

$$x'_j = \frac{1}{\rho} \left[\rho \frac{\rho\sigma_z^2\tau_v\bar{v} - \tau_e\bar{z}}{\rho^2\sigma_z^2 + \tau_e} + \tau_e i_j - \left(\tau_e + \frac{\rho^2\sigma_z^2\tau_v}{\rho^2\sigma_z^2 + \tau_e} \right) p \right]. \quad (5)$$

The only difference between the investment amount of a firm partner and a stand-alone entrepreneur is that the demand of a size- n firm partner has an additional factor of n in front of $\tau_e(i_j - p)$. This means that when in a firm, entrepreneurs become more responsive to her own private information relative to the (imperfectly rationally inferred) public information from equilibrium market price.

Partner j 's expected utility can also be calculated

Corollary 1.3. *Partner j 's expected utility is given by*

$$-\frac{\exp\left(-\frac{1}{2}\frac{1}{\tau_e + \frac{\pi^2}{\gamma^2\sigma_z^2} + \tau_v}\left[\tau_e i_j - \frac{\pi}{\gamma^2\sigma_z^2}(\mu - \gamma\bar{z}) + \tau_v \bar{v} - (\tau_e + \tau_v + \frac{\pi^2}{\gamma^2\sigma_z^2} - \frac{\pi}{\gamma^2\sigma_z^2})p\right]^2\right)}{\sqrt{\frac{n\tau_e + \frac{\pi^2}{\gamma^2\sigma_z^2} + \tau_v}{\tau_e + \frac{\pi^2}{\gamma^2\sigma_z^2} + \tau_v}}} \quad (6)$$

Given Corollary 1.3, we are ready to show the main result of the paper, to be detailed in Subsection 1.3.

1.3 Sharing Revenues Versus Sharing Signals

To demonstrate the *de facto* information aggregation effect of a partnership, and its superiority over direct communication when the latter is subject to either lack of truth-telling incentives or communication cost or delay, I first analyze a hypothetical case in which direct communication is frictionless. The main result, to be summarized in Theorem 1.5, is that even if no communication occurs at all, a partnership gives identical payoffs and thus expected utilities to its partners as if they had enjoyed truthful, costless, and instant direct communication.

When n entrepreneurs enjoy frictionless direct communication prior to making their investment decisions, each of them base her decision on the average of the private signals of all n entrepreneurs. Denote $\frac{1}{n} \sum_{k \text{ within alliance}} i_k$ as $i^* = v + e^*$, then $v \perp\!\!\!\perp e^*$ and

$$e^* \sim \mathcal{N}(0, \frac{1}{n}\tau_e^{-1}).$$

Consider an arbitrary linear valuation system in which $p = \mu + \pi v - \gamma z$. The investment by member j in an signal-sharing alliance is given by maximizing

$$\mathbb{E}[-\exp(-\rho(v-p)x_j|i^*, p)], \text{ and thus} \quad (7)$$

$$\Rightarrow x_j = \frac{\mathbb{E}(v|i^*, p) - p}{\rho \text{Var}(v|i^*, p)} = \frac{1}{\rho} [\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n\tau_e i^* - (n\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2})p] \quad (8)$$

$$\text{(Notice that } \begin{cases} \mathbb{E}(v|i^*, p) &= \frac{\gamma^2 \sigma_z^2 \tau_e i^* + \frac{1}{n} \pi (p - \mu + \gamma \bar{z}) + \frac{1}{n} \gamma^2 \sigma_z^2 \tau_v \bar{v}}{\gamma^2 \sigma_z^2 \tau_e + \frac{1}{n} \pi^2 + \frac{1}{n} \gamma^2 \sigma_z^2 \tau_v} \\ \text{Var}(v|i^*, p) &= \frac{\frac{1}{n} \gamma^2 \sigma_z^2}{\gamma^2 \sigma_z^2 \tau_e + \frac{1}{n} \pi^2 + \frac{1}{n} \gamma^2 \sigma_z^2 \tau_v} \end{cases} \text{)} \quad (9)$$

Compared to the demand from a partner as in (1), the demand with frictionless direct communication differs only in that it replaces the realization of individual signal i_j with the realization of the averaged signal i^* . Because $\frac{1}{n} \sum_{k \text{ within alliance}} i_k = i^*$, equation (8) immediately shows that the total (*ex post*) investment of a partnership firm and that of all entrepreneurs who enjoy frictionless direct communication is identical. Hence equilibrium market price is the same across both structures. In another word, there is an isomorphism in terms of price between direct communication and forming a partnership. This result leads to a similar isomorphism in terms of (*interim*) expected utility, as summarized below.

Lemma 1.4. *The expected utility for entrepreneur j to form an signal-sharing alliance of size n conditional on her private signal is:*

$$-\frac{\exp\left(-\frac{1}{2} \frac{1}{\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v} \left[\tau_e i_j - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + \tau_v \bar{v} - (\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2})p\right]^2\right)}{\sqrt{\frac{n\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}}} \quad (10)$$

Proof. See Appendix. □

Compariing (6) with (10) and by law of iterated expectations we get the following key result of my paper.

Theorem 1.5. *An entrepreneur’s interim expected utilities conditioning on her own private signal are the same in a partnership and under frictionless direct communication.*

Comment: The expected utility equivalence between forming a partnership and direct communication is shown here within a symmetric Hellwig (1980) market. This is meant to simply illustrate that from an information aggregation perspective, a competitive market with dispersed private information is intrinsically unstable. Rational agents will always have incentive to partner with others.

The equivalence result, however, carries over to much broader settings. As long as the economy features the wisdom of crowds, a linear equilibrium within a partnership achieves *de facto* information aggregation.⁶ Furthermore, as long as market participants also perceive the economy of featuring wisdom of crowd, creating a partnership will be a voluntary outcome of market participants’ utility maximization behavior.⁷ The equivalence insight is a general result that holds regardless whether agents make rational expectation inference from the price (thus holds in a partial equilibrium), whether there is heterogeneity among the agents (see Appendix D), and whether the market is competitive or not (see Appendix E).

Because a partnership serves an information aggregation device, the following result follows immediately.

Corollary 1.6. *When the partnership size is adequate small, each partner’s expected utility strictly increases in the partnership size n .*

Proof. When the partnership size is adequate small, change of partnership size has negligible

⁶Although widely true, the wisdom of crowds is occasionally violated due to either behavioral biases (e.g. the above-average effect is observed in some individualist cultures in which people rate themselves as being above average, say, a better driver than the average) or principle-agent problems (client short-termism leading to herding among fund manager).

⁷The perception of the economy featuring wisdom of crowd is one the underpinnings of the entire rational expectation thoughts. As in one the earliest paper on rational expectation, Muth (1961) points out that rational expectation implicitly assume certain “humbleness” among economic agents. However widely accepted assumption might not be universal. For example, many studies assume that economic agents possess over-confidence or agree to disagree (thus violating the Harsanyi principle).

effect on equilibrium price. Taking derivative of (2) w.r.t n leads to

$$= -\frac{\exp(\frac{A}{2B})}{2B^{\frac{5}{2}}}[A'B - AB' - B], \quad (11)$$

while some algebra would show that $A'B - AB' - B =$

$$- \left\{ [\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2}(\mu - \gamma \bar{z}) + \tau_e i_j - (\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2})p] \text{Var}(v|i_j, p) - (\mathbb{E}(v|i_j, p) - p) \right\}^2 \tau_e - (n-1)\tau_e \text{Var}(v|i_j, p) - 1 < 0. \quad \square$$

1.4 Counter-forces against Partnership Formation

Corollary 1.6 might leave the false impression of partnership creation being “the more the merrier”. However, this is not true given the qualification of the partnership size being adequate small. The qualification itself should not bother us much though, as in reality partnership size will necessarily be restricted by natural forces. For example, search frictions (i.e. it takes time to find another entrepreneurs interested in the same project) naturally limits the size of partnerships unless someone actively advertise her partnership intention to find prospective partners.⁸

In the latter case, the size of a partnership will be restricted by both the decline in value of private information and increased price impact. When a non-zero measure of entrepreneurs become *de facto* more informed via joining partnerships, equilibrium price in the market changes. Because when in a firm, entrepreneurs become more responsive to her own private information relative to the (imperfectly rationally inferred) public information from equilibrium market price (Corollary 1.2), the equilibrium price aggregates more private information (i.e., becomes more efficient), lowering the value of private information. Furthermore, when the partnership size is sufficiently large, the absolute value of each entrepreneur’s investment demand enlarges (also Corollary 1.2). The price impact of each partner’s invest-

⁸Search dynamic is a key element in information percolation models that are built on direct communication.

ment is no longer negligible, force them to “shred their order”, further lowering the value of a partnership. These forces are enough to draw the optimal “boundaries of partnership firms”.

To illustrate the first channel, I investigate the price efficiency, investment behavior, and expected utility when *all* entrepreneurs in the continuum join their respective partnerships of finite size n . Because in this case a non-zero measure (indeed a measure one) of entrepreneurs respond more aggressively to their private information when making investment decisions, equilibrium price changes. Yet the finiteness of n still justifies assuming each entrepreneur within their respective partnerships as price-takers.⁹ The results are summarized below.

Corollary 1.7. *When all entrepreneurs in the continuum join their respective partnerships of finite size n , the equilibrium price becomes*

$$\begin{cases} p &= \mu + \pi v - \gamma z \\ \mu &= \frac{\sigma_z^2 \rho^2}{(n\tau_e + \tau_v)\sigma_z^2 \rho^2 + n^2 \tau_e^2} (\tau_v \bar{v} - \frac{n\tau_e}{\rho \sigma_z^2} \bar{z}) \\ \pi &= -\frac{n\tau_e}{\rho} \gamma = \frac{n\tau_e(n\tau_e + \sigma_z^2 \rho^2)}{(n\tau_e + \tau_v)\sigma_z^2 \rho^2 + n^2 \tau_e^2} \\ \gamma &= -\frac{n\tau_e \rho + \sigma_z^2 \rho^3}{(n\tau_e + \tau_v)\sigma_z^2 \rho^2 + n^2 \tau_e^2} \end{cases}, \quad (12)$$

and entrepreneur j 's equilibrium demand is given by

$$\frac{1}{\rho} [\tau_v \bar{v} - n\tau_e (\frac{\tau_v \bar{v} + \rho \bar{z}}{n\tau_e + \sigma_z^2 \rho^2}) + n\tau_e i_j - (n\tau_e + \frac{\rho^2 \sigma_z^2 \tau_v}{\rho^2 \sigma_z^2 + n\tau_e}) p], \quad (13)$$

which gives entrepreneur j 's expected utility as

$$-\frac{\exp\left(-\frac{\rho^2 \sigma_z^2 [-\rho^2 \sigma_z^2 \tau_v \bar{v} + n\rho \tau_e \bar{z} - \tau_e (n\tau_e + \rho^2 \sigma_z^2) i_j + (n\tau_e^2 + \rho^2 \sigma_z^2 \tau_e + \rho^2 \sigma_z^2 \tau_v) p]^2}{2(n\tau_e + \rho^2 \sigma_z^2)^2 (n^2 \tau_e^2 + \rho^2 \sigma_z^2 (\tau_e + \tau_v))}\right)}{\sqrt{\frac{\frac{n^2 \tau_e^2}{\rho^2 \sigma_z^2} + n\tau_e + \tau_v}{\frac{n^2 \tau_e^2}{\rho^2 \sigma_z^2} + \tau_e + \tau_v}}}, \quad (14)$$

⁹Appendix E further relaxes the price-taking assumption.

Proof. Appendix. □

Equation (14) gives an interior optimal n^* .

2 The Firm as a *de facto* Knowledge Aggregator

The analysis above proves that partnership necessarily accompanies a market with dispersed information. As almost all firms are essential payoff-sharing mechanisms among its stakeholders, the above observation suggests a new theory of the firm based on knowledge aggregation. It is worth pointing out that the *de facto* knowledge aggregation role of a firm is likely to be even more relevant in the corporate world than in the financial market. This is reflected both in terms of reducing communication cost and bypassing direct communication as well as the ensuing fear of knowledge stealth.¹⁰

To stay close with the general setup in the previous section, the following section still focuses on firms with CARA stakeholders, linear production technologies, and normal shocks. The firm's production output only varies with the state of nature and all stakeholders' dedication to the firm, but not other production inputs (e.g. physical capital). These specific assumptions highlights the main intuition and keeps tractability, while being extendable to more general settings. My analysis not only attempts to provide a positive theory describing firms, but also means to send a normative message on how internal organizations and compensation design could be structured to take advantage of *de facto* knowledge aggregation.

¹⁰On one hand, conversations, meetings, and discussions take time, at the cost of leisure, actual work, and missing opportunities; misinterpretation and oblivion create additional attrition to communication; “soft” information like haphazard know-hows, amorphous business acumen, and tacit knowledge à la Grant (1996) are simply too hard to codify and impossible to convey; cognitive capacity limits cap the amount of knowledge an individual can possess (see the rational inattention literature as in e.g. Veldkamp (2011)). On the other hand, preventing knowledge stealth is empirically a serious concern, as Bhidé (1994) reports that 71% of the firms included in the Inc 500 (a list of young, fast growing firms) were founded by people who replicated or modified an idea encountered in their previous employment. Theoretically the concern over critical knowledge stealth is also the departure point in Rajan and Zingales (1998).

2.1 A Firm with CARA Partners and Linear Technology

There are two periods and no discounting. A firm’s charter specifies a compensation scheme among its n stakeholders. I focus on the simplest case in which all stakeholders are symmetric, so that upon firm creation in period $t = 0$, the firm charter stipulates that a given stakeholder j will be compensated $\frac{1}{n}$ of the firm’s earning by the end of period $t = 1$.¹¹ Each stakeholder has CARA utility with risk aversion parameter ρ . The firm has a linear production technology $Y = AL$, where Y is the total earning, $A \sim \mathcal{N}(\bar{A}, \tau_A^{-1})$ is a stochastic total-factor productivity (TFP), and L is total labor input.¹² I assume that labor input is additive, so that $L = \sum_{j=1}^n l_j$, where l_j is the labor input contributed by stakeholder j to the firm. Stakeholder j ’s labor input l_j measures her dedication to the firm’s risky project. By assuming a linear, additive production technology, I shutdown any complementarity in stakeholders’ labors which would mechanically favor firm creation.¹³ For the moment I do not consider private cost to each stakeholder’s labor supply. I will comment on the effect of this additional assumption by the end of the section. Each stakeholder’s human capital is modeled as her private knowledge in assessing the stochastic TFP. Such private knowledge takes form as sum of the true TFP plus idiosyncratic noise, i.e., stakeholder j ’s private knowledge $i_j = A + e_j$, where $A \perp e_j$ and $e_j \sim \mathcal{N}(0, \tau_e^{-1})$. It is assumed that the cost of direct communication is prohibitively high. In light of both theoretical prediction for (e.g. [Rajan and Zingales \(2001a\)](#)) and empirical evidence on (e.g. [Rajan and Wulf \(2006\)](#)) human-capital intensive firms, I assume a flat rather than pyramid governance structure. In this case, each

¹¹Notice that when all stakeholders are ex ante symmetric, an equal-split charter is the optimal contract. Indeed, as Section 2.2 will show, equal sharing is an *endogenous* outcome as long as entrepreneurs have homogeneous preference (though not necessarily having homogeneous information precision) and entrepreneurs voluntarily avoid “side-jobs” and work exclusively for the firm. If stakeholders are financially constrained, their equity ownership specified in the firm charter could well be “sweat equities”.

¹²For simplicity I exclude physical capital here, so total-factor productivity (TFP) coincides with labor-factor productivity (LFP). Readers loyal to typical production models can reinterpret A as LFP.

¹³For different modeling purposes, existing literature usually assumes non-separable production technologies, e.g. [Alchian and Demsetz \(1972\)](#). In these models agents’ labor input choices impose (usually positive) externalities on each other. Such externalities can either come from output (e.g. [Kandel and Lazear \(1992\)](#)) or cost (e.g. [Edmans, Goldstein, and Zhu \(2011\)](#)).

stakeholder independently decides how dedicated she is to the firm's project without having to follow instructions from her colleagues.

The n stakeholders in a firm engage in a Nash equilibrium. In particular, entrepreneur j chooses l_j to maximize

$$\mathbb{E} \left[-\exp \left(-\rho \left[\frac{1}{n} A(l_j + \sum_{k \neq j} l_k) \right] \right) | i_j \right], \quad (15)$$

given her (correct) anticipation of other stakeholders' equilibrium dedication l_k , $k \neq j$. The following theorem provides a linear Nash equilibrium solution.

Theorem 2.1. *Given stakeholder j 's problem (20), her equilibrium labor input is given by*

$$l_j = \frac{\tau_A \bar{A}}{\rho} + \frac{n\tau_e}{\rho} i_j, \quad (16)$$

The following lemma turns out to be useful in later analysis:

Lemma 2.2. *If $\begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \right)$, where $(\rho\sigma_1\sigma_2 - 1)^2 > \sigma_1^2\sigma_2^2$ then*

$$\mathbb{E}[e^{xy}] = \frac{\exp \{ (\theta_2^2\sigma_1^2 - 2\rho\theta_1\theta_2\sigma_1\sigma_2 + \theta_1^2\sigma_2^2 + 2\theta_1\theta_2) / (2[(\rho\sigma_1\sigma_2 - 1)^2 - \sigma_1^2\sigma_2^2]) \}}{\sqrt{(\rho\sigma_1\sigma_2 - 1)^2 - \sigma_1^2\sigma_2^2}}.$$

Proof. Standard integration. □

Proof of Theorem 2.1. A linear symmetric equilibrium is given by $l_k = \pi + \gamma i_k$ for some π and γ . Because

$$\begin{bmatrix} -\frac{1}{n}\rho A \\ l_j + \sum_{k \neq j} l_k \end{bmatrix} \Big|_{i_j} \sim$$

$$\mathcal{N} \left(\begin{bmatrix} -\rho \frac{1}{n} \mathbb{E}(A|i_j) \\ l_j + (n-1)\pi + (n-1)\gamma \mathbb{E}(A|i_j) \end{bmatrix}, \begin{bmatrix} \rho^2 \frac{1}{n^2} \text{Var}(A|i_j) & -\rho \frac{1}{n} (n-1)\gamma \text{Var}(A|i_j) \\ -\rho \frac{1}{n} (n-1)\gamma \text{Var}(A|i_j) & (n-1)^2 \gamma^2 \text{Var}(A|i_j) + (n-1)\gamma^2 \tau_e^{-1} \end{bmatrix} \right)$$

by Lemma A.1, entrepreneur j equivalently minimizes

$$\begin{aligned} & \theta_2^2 \rho^2 \frac{1}{n^2} \text{Var}(A|i_j) + 2\theta_1 \theta_2 \rho \frac{1}{n} (n-1)\gamma \text{Var}(A|i_j) + \theta_1^2 [(n-1)^2 \gamma^2 \text{Var}(A|i_j) + (n-1)\gamma^2 \tau_e^{-1}] + 2\theta_1 \theta_2 \\ \xrightarrow{\text{FOC}} & 2\theta_2 \rho^2 \frac{1}{n^2} \text{Var}(A|i_j) + 2\theta_1 \rho \frac{1}{n} (n-1)\gamma \text{Var}(A|i_j) + 2\theta_1 = 0, \\ & \text{where } \theta_1 = -\rho \frac{1}{n} \mathbb{E}(A|i_j) \text{ and } \theta_2 = l_j + (n-1)\pi + (n-1)\gamma \mathbb{E}(A|i_j). \end{aligned}$$

Plugging in $l_j = \pi + \gamma i_j$ leads to

$$[n\pi + \gamma i_j + (n-1)\gamma \mathbb{E}(A|i_j)] \rho \frac{1}{n} \text{Var}(A|i_j) - \rho \frac{1}{n} \mathbb{E}(A|i_j) (n-1)\gamma \text{Var}(A|i_j) - \mathbb{E}(A|i_j) = 0,$$

and matching coefficients renders $l_j = \frac{\tau_A \bar{A}}{\rho} + \frac{n\tau_e}{\rho} i_j$. $\square$

Given no complementarities in labor inputs, one may be tempted to think that entrepreneurs should be indifferent between running a sole proprietorship or taking part in a firm, nor would firm creation affect real allocation. However, Theorem 2.1 shows that a stakeholder becomes more dedicated to the project when in a firm whenever she has positive assessment of the project prospect (i.e., l_j increases with n when $i_j > 0$). Because a stakeholder's assessment are more likely to be positive for a high-value (A) project, creating a firm (rather than keeping multiple sole proprietorships) helps a good project to receive (probablisticly) higher total labor input. Firm creation improves labor allocation.

This result stems from the fact that a firm serves as a *de facto* information aggregation mechanism. This is summarized in the following theorem.

Theorem 2.3. *Given her private project assessment, an entrepreneur's expected utilities*

of participating in an n -stakeholder firm is identical to as if she could obtain other $n - 1$ entrepreneurs' private knowledge without cost while running a sole proprietorship.

The proof is almost identical to that of Theorem 1.5. One direct implication of this result is that firm creation raises entrepreneurs' expected utilities, and thus is a voluntary outcome of economic evolution. An alternative way to understand the *de facto* information aggregation effect comes in the corporate setting recognizes that a firm's charter changes its stakeholders' risk taking incentives (compared to when running as sole-proprietors). In a CARA-normal setting, each stakeholder's labor input decision is given by a mean-variance trading-off. When in a firm, stakeholder j 's compensation partially comes from $\frac{1}{n}$ of the expected value of her contribution bearing only $\frac{1}{n^2}$ of its variance. Her labor input is thus more responsive to her knowledge because the sharing rule in a firm puts her in front of a (as if) less risky project. This effect is reminiscent of what has become conventional wisdom since the seminal work of Markowitz (1952), Sharpe (1964), and Lintner (1965) that proper diversification achieves optimal return-risk trade-off. Firm creation could thus be alternatively interpreted as a mechanism for human-capital diversification.

However, the *de facto* information aggregation channel is different from traditional diversification arguments. First, diversification as in portfolio theory relies on pooling multiple assets and forming portfolios, yet in a revenue-sharing firm model only one single investment project is needed, and the *de facto* information aggregation is achieved through teaming agents.¹⁴ Second, portfolio theory diversification does not require asymmetric information, yet the *de facto* information aggregation channel is built on stakeholders having dispersed private information. In fact, if there is no private knowledge (i.e., $\tau_e = 0$), revenue-sharing would make no difference. Thus there is a fundamental difference between creating a revenue-sharing firm and leaving the capital market laissez-faire.

¹⁴The classic Diamond (1984) is also a diversification story based on pooling assets, although outside of the mean-variance framework.

We shall note that for the results above to hold, it appears necessary to assume that each entrepreneur's precision of information as well as risk aversion are common knowledge. This assumption itself is innocuous, as in reality potential partners in a firm usually either know the others pretty well, or conduct due diligence to assure the other's credential before firm creation. But indeed, the common knowledge requirement on information precision is not essential, as the next section will show.

2.2 Asymmetric knowledge & preference

I extend the above analysis to the scenario where entrepreneurs have heterogeneous preference as well heterogeneous knowledge precision to show the robustness of the *de facto* knowledge aggregation effect of a firm. Still, stakeholder j 's private knowledge $i_j = A + e_j$, where $A \perp e_j$, but idiosyncratic signal precision is assumed to be heterogeneous now: $e_j \sim \mathcal{N}(0, \tau_j^{-1})$. In addition, stakeholder j 's risk-aversion parameter is given by ρ_j . Consider a firm with n stakeholders, in which entrepreneur j 's share is given by a_j , then entrepreneur j chooses l_j to maximize

$$\mathbb{E} \left[-\exp \left(-\rho_j \left[a_j A(l_j + \sum_{k \neq j} l_k) \right] \right) | i_j \right], \quad (17)$$

given her perception of other stakeholders' equilibrium labor input l_k , $k \neq j$. The following theorem provides a linear Nash equilibrium solution for a given stakeholder.

Theorem 2.4. *A linear Nash equilibrium exists only when $a_j = \frac{\frac{1}{\rho_j}}{\sum_{i=1}^n \frac{1}{\rho_i}}$, and stakeholder j 's equilibrium labor input is given by*

$$\frac{\tau_A \bar{A}}{\rho_j} + \left(\sum_{i=1}^n \frac{1}{\rho_i} \right) \tau_j i_j, \quad (18)$$

Proof of Theorem 2.4. A linear symmetric equilibrium is given by $l_k = \pi_k + \gamma_k i_k$ for some

π_k and γ_k . Because

$$\begin{bmatrix} -a_j \rho_j A \\ l_j + \sum_{k \neq j} l_k \end{bmatrix}_{|i_j} \sim$$

$$\mathcal{N} \left(\begin{bmatrix} -\rho_j a_j \mathbb{E}(A|i_j) \\ l_j + \sum_{k \neq j} \pi_k + \sum_{k \neq j} \gamma_k \mathbb{E}(A|i_j) \end{bmatrix}, \begin{bmatrix} \rho_j^2 a_j^2 \text{Var}(A|i_j) & -\rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) \\ -\rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) & (\sum_{k \neq j} \gamma_k)^2 \text{Var}(A|i_j) + \sum_{k \neq j} \gamma_k^2 \tau_k^{-1} \end{bmatrix} \right)$$

Entrepreneur j equivalently minimizes

$$\begin{aligned} & \theta_2^2 \rho_j^2 a_j^2 \text{Var}(A|i_j) + 2\theta_1 \theta_2 \rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) + \theta_1^2 [(\sum_{k \neq j} \gamma_k)^2 \text{Var}(A|i_j) + \sum_{k \neq j} \gamma_k^2 \tau_k^{-1}] + 2\theta_1 \theta_2 \\ \xrightarrow{\text{FOC}} & 2\theta_2 \rho_j^2 a_j^2 \text{Var}(A|i_j) + 2\theta_1 \rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) + 2\theta_1 = 0, \\ & \text{where } \theta_1 = -\rho_j a_j \mathbb{E}(A|i_j) \text{ and } \theta_2 = l_j + \sum_{k \neq j} \pi_k + \sum_{k \neq j} \gamma_k \mathbb{E}(A|i_j). \end{aligned}$$

Plugging in $l_j = \pi_j + \gamma_j i_j$ leads to

$$[\sum_{k \neq j} \pi_k + \pi_j + \gamma_j i_j + \sum_{k \neq j} \gamma_k \mathbb{E}(A|i_j)] \rho_j^2 a_j^2 \text{Var}(A|i_j) - \rho_j a_j \mathbb{E}(A|i_j) \rho a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) - \rho_j a_j \mathbb{E}(A|i_j) = 0$$

and matching coefficients renders $\gamma_j = \frac{\tau_j}{\tau_A \bar{A}} \Pi$, $\frac{\Pi}{\tau_A \bar{A}} \rho_j a_j = 1$ ($\Pi \doteq \sum_{i=1}^n \pi_i$), and thus $a_j = \frac{\frac{1}{\rho_j}}{\sum_{i=1}^n \frac{1}{\rho_i}}$, $\Pi = \left(\sum_{i=1}^n \frac{1}{\rho_i} \right) \tau_A \bar{A}$. Notice that π_k is indeterminate. It is natural to look at one particular equilibrium in which $\pi_k = \frac{\tau_A \bar{A}}{\rho_k}$, leading to $l_j = \frac{\tau_A \bar{A}}{\rho_j} + \left(\sum_{i=1}^n \frac{1}{\rho_i} \right) \tau_j i_j$. $\square$

Notice that if entrepreneur j has full information, her labor supply would be

$$l_j = \frac{\tau_A \bar{A}}{\rho_j} + \frac{\sum_{i=1}^n \tau_i}{\rho_j} i^*, \text{ where } i^* = \frac{\sum_{k=1}^n \tau_k i_k}{\sum_{k=1}^n \tau_k}.$$

Thus entrepreneur j 's payoff is

$$A \left[\frac{\tau_A \bar{A}}{\rho_j} + \frac{\sum_{i=1}^n \tau_i}{\rho_j} l^* \right]$$

under both full information and in a firm. When the entrepreneurs have homogeneous risk preference (but possibly heterogeneous knowledge precisions), a $\frac{1}{n}$ equal sharing rule is optimal.

Side-jobs outside of the firm If a firm disregard each partner's risk preference and stipulates arbitrary sharing rules, its stakeholders will have incentives to conduct "side-job" outside of the firm. To see this, consider an entrepreneur j who has a_j shares in a firm choosing l_j and L_j to maximize

$$\mathbb{E} \left[-\exp \left(-\rho_j \left[a_j A(l_j + \sum_{k \neq j} l_k) + AL_j \right] \right) | i_j \right], \quad (19)$$

given her (correct) anticipation of other stakeholders' equilibrium dedication to the firm l_k , $k \neq j$. The following theorem provides a linear Nash equilibrium solution.

Theorem 2.5. *Each stakeholder's expected utility is maximized when ownership shares in the firm is divided according to stakeholders' risk preferences, i.e. $a_j = \frac{\frac{1}{\rho_j}}{\sum_{i=1}^n \frac{1}{\rho_i}}$. In the resulting linear Nash equilibrium, stakeholders do not conduct "side-jobs" outside of the firm.*

Proof of Theorem 2.5. A linear symmetric equilibrium is given by $l_k + \frac{L_k}{a_k} = \pi_k + \gamma_k l_k + \frac{\Pi_k + \Gamma_k l_k}{a_k}$ for some π_k and γ_k . Because

$$\left[\begin{array}{c} -a_j \rho_j A \\ l_j + \frac{L_j}{a_j} + \sum_{k \neq j} l_k \end{array} \right]_{|i_j} \sim$$

$$\mathcal{N} \left(\begin{bmatrix} -\rho_j a_j \mathbb{E}(A|i_j) \\ l_j + \frac{L_j}{a_j} + \sum_{k \neq j} \pi_k + \sum_{k \neq j} \gamma_k \mathbb{E}(A|i_j) \end{bmatrix}, \begin{bmatrix} \rho_j^2 a_j^2 \text{Var}(A|i_j) & -\rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) \\ -\rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) & (\sum_{k \neq j} \gamma_k)^2 \text{Var}(A|i_j) + \sum_{k \neq j} \gamma_k^2 \tau_k^{-1} \end{bmatrix} \right)$$

by Lemma A.1, entrepreneur j equivalently minimizes

$$\begin{aligned} & \theta_2^2 \rho_j^2 a_j^2 \text{Var}(A|i_j) + 2\theta_1 \theta_2 \rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) + \theta_1^2 [(\sum_{k \neq j} \gamma_k)^2 \text{Var}(A|i_j) + \sum_{k \neq j} \gamma_k^2 \tau_k^{-1}] + 2\theta_1 \theta_2 \\ \xrightarrow{\text{FOC}} & 2\theta_2 \rho_j^2 a_j^2 \text{Var}(A|i_j) + 2\theta_1 \rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) + 2\theta_1 = 0, \end{aligned}$$

$$\text{where } \theta_1 = -\rho_j a_j \mathbb{E}(A|i_j) \text{ and } \theta_2 = l_j + \frac{L_j}{a_j} + \sum_{k \neq j} \pi_k + \sum_{k \neq j} \gamma_k \mathbb{E}(A|i_j).$$

Plugging in $l_j + \frac{L_j}{a_j} = \pi_j + \gamma_j i_j + \frac{\Pi_k + \Gamma_k i_k}{a_k}$ leads to

$$[\sum_{k \neq j} \pi_k + \pi_j + \gamma_j i_j + \frac{\Pi_k + \Gamma_k i_k}{a_k} + \sum_{k \neq j} \gamma_k \mathbb{E}(A|i_j)] \rho_j^2 a_j^2 \text{Var}(A|i_j) - \rho_j a_j \mathbb{E}(A|i_j) \rho_j a_j \sum_{k \neq j} \gamma_k \text{Var}(A|i_j) - \rho_j a_j \mathbb{E}(A|i_j) = 0$$

and matching coefficients renders $\gamma_j + \frac{\Gamma_j}{a_j} = \frac{\tau_j}{\tau_A A} (\Pi + \frac{\Pi_j}{a_j})$, $\frac{1}{\tau_A A} (\Pi + \frac{\Pi_j}{a_j}) \rho_j a_j = 1$ ($\Pi \doteq \sum_{i=1}^n \pi_i$), □

Thus expected utility is given by

$$-\frac{\sqrt{\tau_A + \tau_j} \exp\{-\frac{(\tau_A + \tau_j) \mathbb{E}^2(A|i_j)}{2}\}}{\sqrt{\tau_A + \tau_j + 2\rho_j a_j \sum_{k \neq j} \gamma_k - \rho_j^2 a_j^2 \sum_{k \neq j} \gamma_k^2 \tau_k^{-1}}},$$

which is maximized at $\gamma_j = \frac{\tau_j}{\rho_j a_j}$. Plugging in $\gamma_j + \frac{\Gamma_j}{a_j} = \frac{\tau_j}{\tau_A A} (\Pi + \frac{\Pi_j}{a_j})$ and $\frac{1}{\tau_A A} (\Pi + \frac{\Pi_j}{a_j}) \rho_j a_j = 1$ lead to the at the optimal γ_j , $\Gamma_j = 0$. Thus for any given sharing rule a_k ($k = 1, \dots, n$), there exists a linear equilibrium in which each entrepreneur optimally chooses her amount of labor supply both within and outside of a firm. In particular, when a_j is chosen to be

$\frac{\frac{1}{\rho_j}}{\sum_{i=1}^n \frac{1}{\rho_i}}$, labor supply in the firm can be stipulated so that no entrepreneur has incentive to work outside of the firm.

The incentive to join Given $a_j = \frac{\frac{1}{\rho_j}}{\sum_{i=1}^n \frac{1}{\rho_i}}$, (20) leads to

$$-\frac{\sqrt{\tau_A + \tau_j} \exp \left\{ -\frac{(\tau_A + \tau_j) \mathbb{E}^2(A|i_j)}{2} \right\}}{\sqrt{\tau_A + \sum_{i=1}^n \tau_i}},$$

while entrepreneur j 's utility as a sole-proprietor is given by

$$-\exp \left\{ -\frac{(\tau_A + \tau_j) \mathbb{E}^2(A|i_j)}{2} \right\}.$$

thus entrepreneurs always want to create a firm, even if partnering with those with inferior knowledge. The reason is that any entrepreneur can always change his actions after joining a firm, and those with inferior knowledge still adds a piece of independent information.

Remarks on free-riding A familiar concern over any sharing mechanisms is the moral hazard of free-riding ([Holmström \(1982\)](#)). For example, if endogenous knowledge acquisition is added to the model, any private cost to acquiring knowledge would generate free-riding incentives which counteracts the benefit from *de facto* knowledge aggregation. This problem shall not concern us, though, if our focus is on comparing the relative role of a firm versus the market in *de facto* knowledge aggregation. This is because a similar free-riding problem also plague the financial market with endogenous information acquisition, summarized in [Grossman \(1976\)](#) and [Grossman and Stiglitz \(1980\)](#).

Another type of free-riding problem would arise if an explicit cost to providing labor input is present. In this case, the decision to form a firm involves a trade-off between the benefit from *de facto* knowledge aggregation and a free-riding problem. In general, when

the benefit from *de facto* knowledge aggregation dominates the cost due to free-riding (as in most knowledge intensive firms, e.g. professional service firms), a partnership feature would be more prevalent. This trade-off also helps delineate firm boundary. To illustrate, I analyze a specific example with constant unit labor cost c .

With a unit labor cost c , entrepreneur j chooses l_j to maximize

$$\mathbb{E} \left[-\exp \left(-\rho \left[\frac{1}{n} A(l_j + \sum_{k \neq j} l_k) - cl_j \right] \right) | i_j \right], \quad (20)$$

given her anticipation of other stakeholders' equilibrium labor input level l_k , $k \neq j$. Using similar solution technique, the stakeholder j 's equilibrium dedication to the firm is given by

$$l_j = \frac{\tau_A \bar{A}}{\rho} - \frac{1}{\rho} nc (\tau_A + n\tau_e) + \frac{n\tau_e}{\rho} i_j \quad (21)$$

The equilibrium dedication level consists of three parts. The first part $\frac{\tau_A \bar{A}}{\rho}$ is a constant, the second part $-\frac{1}{\rho} nc (\tau_A + n\tau_e)$ represents a free-riding effect, and the third part $\frac{n\tau_e}{\rho} i_j$ represents the *de facto* knowledge aggregation effect. Under this framework, the relative magnitudes of the two effects determines optimal firm size.

3 General Implications for Corporate Finance

Viewing firms as *de facto* knowledge aggregators sheds new light on several corporate finance topics. Corporate governance, for example, is defined by [Shleifer and Vishny \(1997\)](#) as a study that “deals with the ways in which suppliers of finance to corporations assure themselves of getting a return on their investment”, with “a straightforward agency perspective, sometimes referred to as separation of ownership and control”. The *de facto* knowledge aggregation theory of the firm, however, suggests another possible side of the coin.

In an age of information explosion, business success increasingly asks for knowledge and

talent possessed by not only a single individual but possibly a large team. Compared to physical assets, knowledge and talent are inalienable, rendering a governance theory based on property rights and asset ownership reallocation (e.g. [Grossman and Hart \(1986\)](#)) less applicable. It is then natural to ask what governance structures best glue human capital together, and whether such governance structures are ever superior to pure market allocations.¹⁵ The *de facto* knowledge aggregation view is thus a response to his call for “search of new foundations” (of corporate finance / theory of the firm) in [Zingales \(2000\)](#), emphasizing that existing models are in need of reexamination as traditional asset intensive firms are now being gradually peripheralized by human capital intensive ones. The *de facto* knowledge aggregation view also adds to the rising topic on internal governance of the firm (see e.g. [Acharya, Myers, and Rajan \(2011\)](#) and [Rajan \(2012\)](#)).¹⁶

The partnership perspective in my analysis features a meritocracy-based governance structure. This assumption follows the spirit of [Aghion and Tirole \(1997\)](#), in which formal authority is distinguished from real authority. An agent with formal authority will exercise her power if and only if she acquires the necessary knowledge to so, or otherwise she delegates decision-making to her more knowledgeable subordinates. Aghion and Tirole focus on how the allocation of formal authority alters agents’ *ex ante* knowledge acquisition incentives. In both revenue-sharing models of my paper, entrepreneurs make decisions without others’ interference, as their private knowledge (either about the mapping from the information set to the optimal action or the information set itself) grant them real authority. This decentralized governance structure lines up with the flat organization, teamwork focus, and advocated “workforce democracy” found in most human capital intensive firms. It also differentiates my model from the social choice problem of [Wilson \(1968\)](#).

¹⁵Gluing human capital and preventing talent attrition is an important consideration in modern corporate governance. The consequence of neglecting it is vividly illustrated in the case of the British advertising agency Saatchi and Saatchi documented in [Rajan and Zingales \(2000\)](#).

¹⁶[Berk, Van Binsbergen, and Liu \(2014\)](#) and [Cheng, Massa, Spiegel, and Zhang \(2012\)](#), among others, focus on a particular type of human-capital intensive firms – mutual fund families.

Viewing firms as sharing mechanisms also helps interpret some recent trends in capital structure changes. In a recent discussion on “secular stagnation” among industrialized economies, [Summers \(2014\)](#) pinpoints the “reductions in demand for debt-financed investment”, and contends that “probably to a greater extent, it is a reflection of the changing character of productive economic activity.”¹⁷ Traditional asset-intensive industries are debt-friendly, as assets serve as stable collateral, and allow outside investors to take a passive role in the firm’s operation (except in default).¹⁸ Firms with intensive knowledge inputs, however, require a more active role of all input providers, e.g. the more active roles played by venture capitalists than commercial banks, the adoption of equity-based employee compensation, and little involvement of passive creditors in purely knowledge-based firms (law firms, strategic management (but not IT) consulting firms, etc.) – even though such industry might be most subject to insider moral hazard or unverifiable cash flow, which traditional theories (e.g. [Innes \(1990\)](#) and [Townsend \(1979\)](#)) would predict in favor of debt-financing.¹⁹ In a broader sense, my theory relates to the partnership model of outside equity investors in [Myers \(2000\)](#).

The knowledge aggregation effect of a firm also provide new perspectives on valuation. The nexus of *explicit* contracts view of the firm à la [Alchian and Demsetz \(1972\)](#) and [Jensen and Meckling \(1979\)](#) assumes that the compensation to all stakeholders but shareholders could be explicitly contracted, and thus maximizing shareholder value equates to maximizing social welfare of all firm stakeholders, since equity holders are the only residual rights owner. This view is no long valid once we take into consideration of employee human capital, as

¹⁷Summers elaborates further: “Ponder that the leading technological companies of this age – I think, for example, of Apple and Google – find themselves swimming in cash and facing the challenge of what to do with a very large cash hoard. Ponder the fact that WhatsApp has a greater market value than Sony, with next to no capital investment required to achieve it. Ponder the fact that it used to require tens of millions of dollars to start a significant new venture, and significant new ventures today are seeded with hundreds of thousands of dollars. . .”

¹⁸[Williamson \(1988\)](#) sympathizes this perspective.

¹⁹See also [Berk, Stanton, and Zechner \(2010\)](#) and [Berk and Walden \(2013\)](#) on the implications of human capital on capital structure and asset pricing.

residual rights owners can no longer be summarized as one representative person, and their internal relations do matter. This is particularly true for a knowledge-based firm.²⁰

4 Other Related Literature

The theoretical results of my paper is related to studies on the equilibrium and efficient use of information. In a linear-quadratic setup featuring asymmetric information and strategic complementarity/substitutability, [Angeletos and Pavan \(2007\)](#) show that redistribution among individuals can achieve efficient information aggregation as an equilibrium outcome. In my profit-sharing setup, I obtain a similar result, although strategic complementarity/substitutability are not present.

The indirect information transmission with profit-sharing can also be linked to a series of papers by on indirect information sale in the financial market.²¹ What Admati and Pfleiderer mean by indirect sale of information is that when informed investors manage delegated portfolios on behalf of those uninformed for a fee, they are essentially selling their information. Following this logic, when information in the economic system is dispersed, investors would have incentives to delegate their wealth to each other. The result of this mutual delegation is exactly the profit-sharing mechanism seen in a firm. Admati and Pfleiderer uses their indirect information sale insight to explain the rise of institutional investors. In this sense, we can also view their results as a specific yet important example for a more general theory of the firm.²²

Third, from a purely technical perspective, a cooperative provides a game-theoretical implementation of a rational expectation equilibrium. So my result is connected to the im-

²⁰Several recent papers have investigated the valuation implication of firm's non-tangible assets. For example, [Eisfeldt and Papanikolaou \(2013\)](#) document that firms with more organization capital have average returns that are 4.6% higher than firms with less organization capital. [Zhang \(2014\)](#) studies the implications of employee's limited commitment to the firm on cash flow volatility.

²¹See [Admati and Pfleiderer \(1990\)](#), etc.

²²Also see [García and Vanden \(2009\)](#) on wealth delegation with endogenous information acquisition.

plementation theory literature in mechanism design, whose focus is on designing mechanisms to achieve one equilibrium outcome via another equilibrium concept. [Palfrey \(2002\)](#) provides a nice introduction to implementation theory, while [Blume and Easley \(1990\)](#) studies implementation of Walrasian expectations equilibrium in a general setting.

The second part of my paper also contributes the literature on the theory of the firm. Over 70 years' academic endeavors on this fundamental topic makes an exhaustive reference a daunting job, so I only attempt to classify some well-known contributions around several major lines of thoughts and connect them with the revenue-sharing theory.

The neoclassic theory views firms simply as production technology sets, and firms *per se* are void of meaningful definitions. The first and foremost question, raised by the seminal work of [Coase \(1937\)](#), asks what essentially defines a firm, and how within-firm organization is distinguished from market contracting. Coase identifies authority, which is useful when contracting is costly, as the defining feature that differs within-firm transactions apart from market contracting. Two questions remain to be answered though in Coase's original argument, the first being what constitutes contracting costs, and the second being a formal definition of authority.

On contracting costs, [Williamson \(1975\)](#), [Klein, Crawford, and Alchian \(1978\)](#), [Williamson \(1979\)](#), and [Williamson \(1985\)](#) identifies *ex post* haggling as a source of cost to contracting. In my humble opinion, another important friction to contracting lies in the nonexistence of Pareto optimal, incentive compatible, and budget-balancing bilateral bargaining outcomes under two-sided asymmetric information ([Myerson and Satterthwaite \(1983\)](#)), although to my best knowledge no resolutions have yet been proposed in this direction.

The formalization of authority spearheads the development of the incomplete contracting approach. Under this theme, [Grossman and Hart \(1986\)](#) and [Hart and Moore \(1990\)](#) argue that asset ownership determines the allocation of residual rights (or authority). In this property rights theory of the firm, physical assets play vital roles for the very existence of

firms as they entangle other production inputs around it and give birth to firms.²³ Hart (1995) even goes further and argues that “a firm’s non-human assets, then, simply represent the glue that keeps the firm together . . . If non-human assets do not exist, then it is not clear what keeps the firm together” (p. 57).²⁴ However, as human capital intensive firms arise in the knowledge economy, Rajan and Zingales (1998) point out the narrowness of property rights theory and develop a new theory based on “access” to critical resources.

The Coase authority-cost paradigm is not the only framework for understanding firms. For example, the “nexus of contracts” theory views a firm as a legal illusion no more than a central contracting party to subsume a complex of multilateral contracts (Alchian and Demsetz (1972)). Jensen and Meckling (1979) specifically focus on the principal-agent contracting problem between a firm owner and the management subject to moral hazard. In the market intermediary theory, Spulber (1999) interprets firms as centralized exchanges to reduce market search costs. A few papers take a knowledge perspective on the essence of the firm. Demsetz (1988) emphasizes information cost reduction as a foundation of firms, and Grant (1996) proposes a knowledge-based theory of the firm. However, both papers take information cost reduction and knowledge aggregation with a firm as given, without concrete, micro-founded mechanisms supporting the effects. Bolton and Dewatripont (1994) analyze how organizations minimize information processing and communicating costs,²⁵ their study, however, abstracts from firm stakeholders’ communication incentives, which is emphasized in my revenue-sharing theory of the firm.²⁶

My paper attempts to add to the general discussions on the theory of the firm, with specific relevance in our current knowledge economy. This specific focus is also found in

²³See Hart (1989) and Holmström and Tirole (1989) for reviews.

²⁴This view is partially supported by Kaplan, Sensoy, and Strömberg (2009).

²⁵See also Winter (2006), Winter (2010).

²⁶On how information distribution among stakeholders influence firm governance, Dicks and Fulghieri (2014) show how ambiguity aversion generates endogenous disagreement between owners and managers, which in turn determines optimal governance structure.

Rajan and Zingales (2001b), who espouse the “critical resource theory” for human-capital intensive firms; in Zingales (2000), who connects this trend to the change of landscape in corporate finance studies; and in the AFA presidential address of Rajan (2012), who further highlights the linkage between firm organization and corporate finance in the context of the life-cycle of entrepreneurial firms.

5 Conclusion

One lesson that the studies on rational expectation equilibria teaches us is that the market could be viewed as an information aggregation device. The current paper points out that a partnership structure could also achieve this information aggregation effect. Because of the noises intrinsic to a large market, and because of the incentive problems and costs associated with other non-market information aggregation mechanism, in comparison the partnership structure demonstrates many superiority. Under this framework, a competitive market does not exist in isolation when investors have rational expectation. Firms arise as endogenous outcomes out of a market economy. This result sheds new light on the theory of the firm.

By focusing on the *de facto* knowledge aggregation effect, the payoff-sharing partnerships interpretation abstracts from other channels that could also justify the endogenous arise of firms in a market economy. In a separate paper, I derive how a payoff-sharing partnerships overcomes market failures caused by the famous Hirshleifer (1971) effect. The role of a partnership in encouraging truthful direct communication is also only discussed in passing. This unfavorable treatment by no means downplays the importance of the truth-telling encouragement channel, rather just because the underlying intuition is somewhat straightforward.

The *de facto* knowledge aggregation intuition also has implications that extends beyond corporate finance. It could potentially describe the rise of “too-big-to-fail” banks, the role of mutual fund families, and provide an incentive-compatibility alternative to information

percolation. I leave these application to a separate paper.

Last but not least, the equivalence between payoff-sharing and sharing information might suggest an often neglected *de facto* information channel. For example, how effective are Chinese walls between firm subdivisions that are exposed to conflict of interests? Does profit-sharing camouflage insider trading? These questions might concern regulators, and more thorough development is scheduled for future research.

Appendix

A Proof of Theorem 1.1

The following lemma turns out to be useful in later analysis:

Lemma A.1. *If $\begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right)$, where $(\rho\sigma_1\sigma_2 - 1)^2 > \sigma_1^2\sigma_2^2$ then*

$$\mathbb{E}[e^{xy}] = \frac{\exp\{(\theta_2^2\sigma_1^2 - 2\rho\theta_1\theta_2\sigma_1\sigma_2 + \theta_1^2\sigma_2^2 + 2\theta_1\theta_2)/[2((\rho\sigma_1\sigma_2 - 1)^2 - \sigma_1^2\sigma_2^2)]\}}{\sqrt{(\rho\sigma_1\sigma_2 - 1)^2 - \sigma_1^2\sigma_2^2}}.$$

Proof. Standard integration. □

Proof of Theorem 1.1: Entrepreneur j 's investment in the risky project when in a firm is given by maximizing

$$\mathbb{E}\left[-\exp(-\rho(v-p)(x_j + \sum_{k \neq j} x_k)/n)|i_j, p\right] \quad (22)$$

Focusing on symmetric linear equilibria, assume $x_k = n\alpha_0 + n\alpha_1 i_k + n\alpha_2 p$, where α_i ($i = 0, 1, 2$) are functions of n . Notice that

$$(x_j + \sum_{k \neq j} x_k)/n = x_j/n + (n-1)\alpha_0 + (n-1)\alpha_1 v + (n-1)\alpha_2 p + \alpha_1 \sum_{k \neq j} e_k, \quad (23)$$

thus

$$\left(\begin{array}{c} -\rho(v-p) \\ (x_j + \sum_{k \neq j} x_k)/n \end{array}\right)_{|i_j, p} \sim \mathcal{N}\left(\begin{array}{c} \begin{bmatrix} -\rho\mathbb{E}(v|i_j, p) + \rho p \\ x_j/n + (n-1)\alpha_0 + (n-1)\alpha_1\mathbb{E}(v|i_j, p) + (n-1)\alpha_2 p \end{bmatrix} \\ \begin{bmatrix} \rho^2\text{Var}(v|i_j, p) & -\rho(n-1)\alpha_1\text{Var}(v|i_j, p) \\ -\rho(n-1)\alpha_1\text{Var}(v|i_j, p) & (n-1)^2\alpha_1^2\text{Var}(v|i_j, p) + \alpha_1^2(n-1)\tau_e^{-1} \end{bmatrix} \end{array}\right) \quad (24)$$

By Lemma A.1, the certainty equivalent of (42) is

$$-\frac{\exp(\frac{A}{2B})}{\sqrt{B}}, \quad (25)$$

where

$$\begin{aligned} A &= (-\rho\mathbb{E}(v|i_j, p) + \rho p)^2[(n-1)^2\alpha_1^2\text{Var}(v|i_j, p) + \alpha_1^2(n-1)\tau_e^{-1}] \\ &\quad + [x_j/n + (n-1)\alpha_0 + (n-1)\alpha_1\mathbb{E}(v|i_j, p) + (n-1)\alpha_2p]^2\rho^2\text{Var}(v|i_j, p) \\ &\quad + 2(-\rho\mathbb{E}(v|i_j, p) + \rho p)[x_j/n + (n-1)\alpha_0 + (n-1)\alpha_1\mathbb{E}(v|i_j, p) + (n-1)\alpha_2p][1 + \\ &\quad \quad \rho(n-1)\alpha_1\text{Var}(v|i_j, p)] \\ B &= [\rho(n-1)\alpha_1\text{Var}(v|i_j, p) + 1]^2 \\ &\quad - \rho^2\text{Var}(v|i_j, p)[(n-1)^2\alpha_1^2\text{Var}(v|i_j, p) + \alpha_1^2(n-1)\tau_e^{-1}] \end{aligned}$$

Taking FOC w.r.t. x_j we get

$$x_j/n + (n-1)\alpha_0 + (n-1)\alpha_1\mathbb{E}(v|i_j, p) + (n-1)\alpha_2p \quad (26)$$

$$= \frac{(-\rho(n-1)\alpha_1\text{Var}(v|i_j, p) - 1)(-\rho\mathbb{E}(v|i_j, p) + \rho p)}{\rho^2\text{Var}(v|i_j, p)} \quad (27)$$

$$= \frac{[\rho(n-1)\alpha_1\text{Var}(v|i_j, p) + 1](\mathbb{E}(v|i_j, p) - p)}{\rho\text{Var}(v|i_j, p)}. \quad (28)$$

Given (3), we have

$$\begin{cases} \mathbb{E}(v|i_j, p) &= \bar{v} + \frac{\gamma^2\sigma_z^2\tau_v^{-1}(i_j - \bar{v}) + \pi\tau_v^{-1}\tau_e^{-1}(p - \mu - \pi\bar{v} + \gamma\bar{z})}{\gamma^2\sigma_z^2\tau_v^{-1} + \pi^2\tau_v^{-1}\tau_e^{-1} + \gamma^2\sigma_z^2\tau_e^{-1}} = \frac{\gamma^2\sigma_z^2\tau_v^{-1}i_j + \pi\tau_v^{-1}\tau_e^{-1}(p - \mu + \gamma\bar{z}) + \gamma^2\sigma_z^2\tau_e^{-1}\bar{v}}{\gamma^2\sigma_z^2\tau_v^{-1} + \pi^2\tau_v^{-1}\tau_e^{-1} + \gamma^2\sigma_z^2\tau_e^{-1}} \\ \text{Var}(v|i_j, p) &= \tau_v^{-1} - \frac{\gamma^2\sigma_z^2\tau_v^{-2} + \pi^2\tau_v^{-2}\tau_e^{-1}}{\gamma^2\sigma_z^2\tau_v^{-1} + \pi^2\tau_v^{-1}\tau_e^{-1} + \gamma^2\sigma_z^2\tau_e^{-1}} = \frac{\gamma^2\sigma_z^2\tau_e^{-1}\tau_v^{-1}}{\gamma^2\sigma_z^2\tau_v^{-1} + \pi^2\tau_v^{-1}\tau_e^{-1} + \gamma^2\sigma_z^2\tau_e^{-1}} \end{cases} \quad (29)$$

In equilibrium $x_j = n\alpha_0 + n\alpha_1i_j + n\alpha_2p$, and Equation (28) leads to

$$n\alpha_0 + \alpha_1i_j + (n-1)\alpha_1\mathbb{E}(v|i_j, p) + n\alpha_2p = \frac{[\rho(n-1)\alpha_1\text{Var}(v|i_j, p) + 1](\mathbb{E}(v|i_j, p) - p)}{\rho\text{Var}(v|i_j, p)}, \quad (30)$$

thus

$$\rho\text{Var}(v|i_j, p)[n\alpha_0 + \alpha_1i_j + n\alpha_2p] = -\rho(n-1)\alpha_1\text{Var}(v|i_j, p)p + \mathbb{E}(v|i_j, p) - p \quad (31)$$

Plug in (29),

$$\rho[n\alpha_0 + \alpha_1i_j + n\alpha_2p] = -\rho(n-1)\alpha_1p - \frac{\gamma^2\sigma_z^2\tau_v^{-1} + \pi^2\tau_v^{-1}\tau_e^{-1} + \gamma^2\sigma_z^2\tau_e^{-1}}{\gamma^2\sigma_z^2\tau_e^{-1}\tau_v^{-1}}p + \quad (32)$$

$$\frac{\gamma^2\sigma_z^2\tau_v^{-1}i_j + \pi\tau_v^{-1}\tau_e^{-1}(p - \mu + \gamma\bar{z}) + \gamma^2\sigma_z^2\tau_e^{-1}\bar{v}}{\gamma^2\sigma_z^2\tau_e^{-1}\tau_v^{-1}} \quad (33)$$

Equalizing coefficients:

$$\begin{cases} \rho n \alpha_0 &= \tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) \\ \rho \alpha_1 &= \tau_e \\ \rho n \alpha_2 &= -(n-1)\tau_e - (\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2}) + \frac{\pi}{\gamma^2 \sigma_z^2} \end{cases} \quad (34)$$

Thus

$$x_j = \frac{1}{\rho} [\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n \tau_e i_j - (n \tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p] \quad (35)$$

Plug (34) in (45) shows that

$$\begin{aligned} A &= (\mathbb{E}(v|i_j, p) - p)^2 [(n-1)^2 \tau_e^2 \text{Var}(v|i_j, p) + \tau_e (n-1)] \\ &+ [\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + \tau_e i_j + (n-1)\tau_e \mathbb{E}(v|i_j, p) - (n \tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p]^2 \text{Var}(v|i_j, p) \\ &- 2(\mathbb{E}(v|i_j, p) - p) [\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + \tau_e i_j + (n-1)\tau_e \mathbb{E}(v|i_j, p) - (n \tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p] \\ &\times [1 + (n-1)\tau_e \text{Var}(v|i_j, p)] \\ B &= 1 + (n-1)\tau_e \text{Var}(v|i_j, p), \text{ and} \end{aligned}$$

$$\begin{aligned} \mathbb{E}(v|i_j, p) &= \frac{\gamma^2 \sigma_z^2 \tau_e i_j + \pi(p - \mu + \gamma \bar{z}) + \gamma^2 \sigma_z^2 \tau_v \bar{v}}{\gamma^2 \sigma_z^2 \tau_e + \pi^2 + \gamma^2 \sigma_z^2 \tau_v} \\ \text{Var}(v|i_j, p) &= \frac{\gamma^2 \sigma_z^2}{\gamma^2 \sigma_z^2 \tau_e + \pi^2 + \gamma^2 \sigma_z^2 \tau_v}. \end{aligned}$$

Futher simplification gives the expression for the expected utility of entrepreneur j . $\square$

B Proof of Corollary 1.7

Proof of Corollary 1.7: When all entrepreneurs create firms, in a linear equilibrium, project valuation and entrepreneur investment still follows

$$\tilde{p} = \mu + \pi \tilde{v} - \gamma \tilde{z} \quad (36)$$

$$x_k = n \alpha_0 + n \alpha_1 i_k + n \alpha_2 p, \text{ for entrepreneur } k, \quad (37)$$

where μ, π, γ and α_i ($i = 0, 1, 2$) are all functions of n to be determined, and by the same arguments in the proof of Corollary 1.3, these coefficients satisfy (34).

Integrate each individual entrepreneurs' investment over the continuum and by market clearing,

$$n \alpha_0 + n \alpha_1 v + n \alpha_2 (\mu + \pi v - \gamma z) + z = 0, \text{ thus} \quad (38)$$

$$\begin{cases} \alpha_0 &= -\alpha_2 \mu \\ \alpha_1 &= -\alpha_2 \pi \\ \alpha_2 &= \frac{1}{n \gamma} \end{cases} \quad (39)$$

Plug in (34) and re-arrange terms,

$$\begin{cases} \mu &= \frac{\sigma_z^2 \rho^2}{(n\tau_e + \tau_v)\sigma_z^2 \rho^2 + n^2 \tau_e^2} (\tau_v \bar{v} - \frac{n\tau_e}{\rho \sigma_z^2} \bar{z}) \\ \pi &= -\frac{n\tau_e}{\rho} \gamma = \frac{n\tau_e(n\tau_e + \sigma_z^2 \rho^2)}{(n\tau_e + \tau_v)\sigma_z^2 \rho^2 + n^2 \tau_e^2} \\ \gamma &= -\frac{n\tau_e \rho + \sigma_z^2 \rho^3}{(n\tau_e + \tau_v)\sigma_z^2 \rho^2 + n^2 \tau_e^2} \end{cases} \quad (40)$$

Plug in (53) and (45) and we get the optimal demand and expected payoff given in (13). $\square$

C Proof of Lemma 1.4

Proof of Lemma 1.4. Once entered into a signal-sharing alliance, entrepreneur j 's expected utility is given by

$$-\exp \left\{ - \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n\tau_e i^* - (n\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right] [\mathbb{E}(v|i^*, p) - p] + \frac{1}{2} \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n\tau_e i^* - (n\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right]^2 \text{Var}(v|i^*, p) \right\} \quad (41)$$

Thus her expected utility before entering the alliance (but after receiving i_j) is

$$\begin{aligned} & -\mathbb{E} \left\{ \exp \left(- \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n\tau_e i^* - (n\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right] [\mathbb{E}(v|i^*, p) - p] \right. \right. \\ & \quad \left. \left. + \frac{1}{2} \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n\tau_e i^* - (n\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right]^2 \text{Var}(v|i^*, p) \right) \middle| i_j, p \right\} \\ = & -\mathbb{E} \left\{ \exp \left(- \frac{1}{2} \frac{1}{n\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v} \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n\tau_e i^* - (n\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right]^2 \right) \middle| i_j, p \right\} \\ = & -\frac{\exp \left(- \frac{1}{2} \frac{1}{\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v} \left[\tau_e i_j - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + \tau_v \bar{v} - (\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right]^2 \right)}{\sqrt{\frac{n\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}}} \end{aligned}$$

The first equation is from (9), while the second one uses the following two facts:

1. If $x \sim \mathcal{N}(\mu, \sigma^2)$, $\mathbb{E}[e^{Ax^2}] = \frac{\exp\left(\frac{A\mu^2}{1-2A\sigma^2}\right)}{\sqrt{1-2A\sigma^2}}$;
2. $\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + n\tau_e i^* - (n\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \big|_{i_j, p} \sim \mathcal{N} \left(\frac{n\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v} \left[\tau_e i_j - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + \tau_v \bar{v} - (\tau_e + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right], \frac{n\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_e + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v} (n-1)\tau_e \right)$

$\square$

D Rational Expectation Equilibrium with Asymmetry Investors

Consider a market in which investors have heterogeneous risk preference and information precision. When n entrepreneurs agree to partner, partner j 's investment in the risky project maximizes

$$\mathbb{E} \left[-\exp(-\rho_j a_j (v - p)(x_j + \sum_{k \neq j} x_k)) | i_j, p \right] \quad (42)$$

Focusing on symmetric linear equilibria, assume $x_k = \alpha_{0k} + \alpha_{1k} i_k + \alpha_{2k} p$. Notice that

$$x_j + \sum_{k \neq j} x_k = x_j + \sum_{k \neq j} \alpha_{0k} + \sum_{k \neq j} \alpha_{1k} v + \sum_{k \neq j} \alpha_{2k} p + \sum_{k \neq j} \alpha_{1k} e_k, \quad (43)$$

thus

$$\begin{pmatrix} -\rho_j a_j (v - p) \\ x_j + \sum_{k \neq j} x_k \end{pmatrix}_{|i_j, p} \sim \mathcal{N} \left(\begin{bmatrix} -\rho_j a_j (\mathbb{E}(v|i_j, p) - p) \\ x_j + \sum_{k \neq j} \alpha_{0k} + \sum_{k \neq j} \alpha_{1k} \mathbb{E}(v|i_j, p) + \sum_{k \neq j} \alpha_{2k} p \\ \rho_j^2 a_j^2 \text{Var}(v|i_j, p) & -\rho_j a_j \sum_{k \neq j} \alpha_{1k} \text{Var}(v|i_j, p) \\ -\rho_j a_j \sum_{k \neq j} \alpha_{1k} \text{Var}(v|i_j, p) & (\sum_{k \neq j} \alpha_{1k})^2 \text{Var}(v|i_j, p) + \sum_{k \neq j} \alpha_{1k}^2 \tau_k^{-1} \end{bmatrix} \right) \quad (44)$$

By Lemma A.1, the certainty equivalent of (42) is

$$-\frac{\exp(\frac{A}{2B})}{\sqrt{B}}, \quad (45)$$

where

$$\begin{aligned} A &= \rho_j^2 a_j^2 (\mathbb{E}(v|i_j, p) - p)^2 \left[\left(\sum_{k \neq j} \alpha_{1k} \right)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j} \alpha_{1k}^2 \tau_k^{-1} \right] \\ &+ \left[x_j + \sum_{k \neq j} \alpha_{0k} + \sum_{k \neq j} \alpha_{1k} \mathbb{E}(v|i_j, p) + \sum_{k \neq j} \alpha_{2k} p \right]^2 \rho_j^2 a_j^2 \text{Var}(v|i_j, p) \\ &- 2\rho_j a_j (\mathbb{E}(v|i_j, p) - p) \left[x_j + \sum_{k \neq j} \alpha_{0k} + \sum_{k \neq j} \alpha_{1k} \mathbb{E}(v|i_j, p) + \sum_{k \neq j} \alpha_{2k} p \right] \left[1 + \rho_j a_j \sum_{k \neq j} \alpha_{1k} \text{Var}(v|i_j, p) \right] \\ B &= \left[1 + \rho_j a_j \sum_{k \neq j} \alpha_{1k} \text{Var}(v|i_j, p) \right]^2 \\ &- \rho_j^2 a_j^2 \text{Var}(v|i_j, p) \left[\left(\sum_{k \neq j} \alpha_{1k} \right)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j} \alpha_{1k}^2 \tau_k^{-1} \right] \end{aligned}$$

Taking FOC w.r.t. x_j we get

$$[x_j + \sum_{k \neq j} \alpha_{0k} + \sum_{k \neq j} \alpha_{1k} \mathbb{E}(v|i_j, p) + \sum_{k \neq j} \alpha_{2k} p] \rho_j a_j \text{Var}(v|i_j, p) \quad (46)$$

$$= (\mathbb{E}(v|i_j, p) - p)[1 + \rho_j a_j \sum_{k \neq j} \alpha_{1k} \text{Var}(v|i_j, p)]. \quad (47)$$

Given (3), we have

$$\begin{cases} \mathbb{E}(v|i_j, p) &= \bar{v} + \frac{\gamma^2 \sigma_z^2 \tau_v^{-1} (i_j - \bar{v}) + \pi \tau_v^{-1} \tau_j^{-1} (p - \mu - \pi \bar{v} + \gamma \bar{z})}{\gamma^2 \sigma_z^2 \tau_v^{-1} + \pi^2 \tau_v^{-1} \tau_j^{-1} + \gamma^2 \sigma_z^2 \tau_j^{-1}} = \frac{\gamma^2 \sigma_z^2 \tau_v^{-1} i_j + \pi \tau_v^{-1} \tau_j^{-1} (p - \mu + \gamma \bar{z}) + \gamma^2 \sigma_z^2 \tau_j^{-1} \bar{v}}{\gamma^2 \sigma_z^2 \tau_v^{-1} + \pi^2 \tau_v^{-1} \tau_j^{-1} + \gamma^2 \sigma_z^2 \tau_j^{-1}} \\ \text{Var}(v|i_j, p) &= \tau_v^{-1} - \frac{\gamma^2 \sigma_z^2 \tau_v^{-2} + \pi^2 \tau_v^{-2} \tau_j^{-1}}{\gamma^2 \sigma_z^2 \tau_v^{-1} + \pi^2 \tau_v^{-1} \tau_j^{-1} + \gamma^2 \sigma_z^2 \tau_j^{-1}} = \frac{\gamma^2 \sigma_z^2 \tau_j^{-1} \tau_v^{-1}}{\gamma^2 \sigma_z^2 \tau_v^{-1} + \pi^2 \tau_v^{-1} \tau_j^{-1} + \gamma^2 \sigma_z^2 \tau_j^{-1}} \end{cases} \quad (48)$$

In equilibrium $x_j = \alpha_{0j} + \alpha_{1j} i_j + \alpha_{2j} p$, and Equation (47) leads to

$$\alpha_{1j} i_j + \sum_{k=1}^n \alpha_{0k} + \sum_{k \neq j} \alpha_{1k} \mathbb{E}(v|i_j, p) + \sum_{k=1}^n \alpha_{2k} p = \frac{(\mathbb{E}(v|i_j, p) - p)[1 + \rho_j a_j \sum_{k \neq j} \alpha_{1k} \text{Var}(v|i_j, p)]}{\rho_j a_j \text{Var}(v|i_j, p)}, \quad (49)$$

thus

$$\rho_j a_j \text{Var}(v|i_j, p) \left[\alpha_{1j} i_j + \sum_{k=1}^n \alpha_{0k} + \sum_{k=1}^n \alpha_{2k} p \right] = \mathbb{E}(v|i_j, p) - p - \rho_j a_j \sum_{k \neq j} \alpha_{1k} \text{Var}(v|i_j, p) p \quad (50)$$

Plug in (48),

$$\begin{aligned} & \rho_j a_j \gamma^2 \sigma_z^2 \tau_j^{-1} \tau_v^{-1} \left[\alpha_{1j} i_j + \sum_{k=1}^n \alpha_{0k} + \sum_{k=1}^n \alpha_{2k} p \right] = \gamma^2 \sigma_z^2 \tau_v^{-1} i_j + \pi \tau_v^{-1} \tau_j^{-1} (p - \mu + \gamma \bar{z}) + \gamma^2 \sigma_z^2 \tau_j^{-1} \bar{v} \\ & - (\gamma^2 \sigma_z^2 \tau_v^{-1} + \pi^2 \tau_v^{-1} \tau_j^{-1} + \gamma^2 \sigma_z^2 \tau_j^{-1}) p - \rho_j a_j \sum_{k \neq j} \alpha_{1k} \gamma^2 \sigma_z^2 \tau_j^{-1} \tau_v^{-1} p \end{aligned}$$

Equalizing coefficients:

$$\begin{cases} \rho_j a_j \sum_{k=1}^n \alpha_{0k} &= \tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) \\ \rho_j a_j \alpha_{1j} &= \tau_j \\ \rho_j a_j \sum_{k=1}^n \alpha_{2k} &= -(\tau_j + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2}) + \frac{\pi}{\gamma^2 \sigma_z^2} - \rho_j a_j \sum_{k \neq j} \frac{\tau_k}{\rho_k a_k} \end{cases} \quad (51)$$

$\Rightarrow$

$$\begin{cases} a_j &= \frac{\frac{1}{\rho_j}}{\sum_{k=1}^n \frac{1}{\rho_k}} \\ \sum_{k=1}^n \alpha_{0k} &= \left(\sum_{k=1}^n \frac{1}{\rho_k} \right) \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) \right] \\ \alpha_{1j} &= \left(\sum_{k=1}^n \frac{1}{\rho_k} \right) \tau_j \\ \sum_{k=1}^n \alpha_{2k} &= \left(\sum_{k=1}^n \frac{1}{\rho_k} \right) \left[-(\sum_{k=1}^n \tau_k + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2}) + \frac{\pi}{\gamma^2 \sigma_z^2} \right] \end{cases} \quad (52)$$

Thus a natural (however not necessarily unique) result is to have

$$x_j = \frac{1}{\rho_j} \left[\tau_v \bar{v} - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) \right] + \left(\sum_{k=1}^n \frac{1}{\rho_k} \right) \tau_j i_j - \frac{1}{\rho_j} \left[\sum_{k=1}^n \tau_k + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2} \right] p \quad (53)$$

Plug (34) in (45) shows that

$$\begin{aligned} \mathbb{E}(v|i_j, p) &= \frac{\tau_j i_j + \frac{\pi}{\gamma^2 \sigma_z^2} (p - \mu + \gamma \bar{z}) + \tau_v \bar{v}}{\tau_j + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v} \\ \text{Var}(v|i_j, p) &= \frac{1}{\tau_j + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}. \\ A &= -(\mathbb{E}(v|i_j, p) - p)^2 \left(\sum_{k=1}^n \tau_k + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v \right) \\ B &= \frac{\sum_{k=1}^n \tau_k + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_j + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}, \end{aligned}$$

thus the expression for the expected utility of entrepreneur j is given by

$$\begin{aligned} & \exp \left(- \frac{(\mathbb{E}(v|i_j, p) - p)^2 (\tau_j + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v)}{2} \right) \\ & - \frac{\sqrt{\frac{\sum_{k=1}^n \tau_k + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_j + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}}}{\sqrt{\frac{\sum_{k=1}^n \tau_k + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_j + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}}} \\ & = - \frac{\exp \left(- \frac{1}{2} \frac{1}{\tau_j + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v} \left[\tau_j i_j - \frac{\pi}{\gamma^2 \sigma_z^2} (\mu - \gamma \bar{z}) + \tau_v \bar{v} - (\tau_j + \tau_v + \frac{\pi^2}{\gamma^2 \sigma_z^2} - \frac{\pi}{\gamma^2 \sigma_z^2}) p \right]^2 \right)}{\sqrt{\frac{\sum_{k=1}^n \tau_k + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}{\tau_j + \frac{\pi^2}{\gamma^2 \sigma_z^2} + \tau_v}}} \end{aligned}$$

Notice that this is identical to the expected utility achieved under complete information (but no partnership creation).

E Rational Expectation Equilibrium with Asymmetric Investors and Imperfect Competition

First consider a heterogeneous agents extension to [Kyle \(1989\)](#). In this case there are N investors who do not form partnerships. Perceiving a residual supply curve $p = p_j + \lambda_j x_j$, investor j chooses

x_j to maximize

$$\begin{aligned}
& \mathbb{E}[-\exp(-\rho_j(v - p(x_j))x_j)|i_j, p], \text{ or} \\
& \mathbb{E}[-\exp(-\rho_j(v - p_j - \lambda_j x_j)x_j)|i_j, p_j] \\
\Leftrightarrow & -\exp\left[-\rho_j(\mathbb{E}(v|i_j, p_j) - p_j)x_j + \rho_j \lambda_j x_j^2 + \frac{1}{2}\rho_j^2 \text{Var}(v|i_j, p_j)x_j^2\right] \\
\stackrel{\text{FOC}}{\implies} & x_j = \frac{(\mathbb{E}(v|i_j, p_j) - p_j)}{2\lambda_j + \rho_j \text{Var}(v|i_j, p_j)}
\end{aligned}$$

thus

$$p = p_j + \lambda_j x_j \quad (54)$$

$$= p_j + \frac{\lambda_j(\mathbb{E}(v|i_j, p_j) - p_j)}{2\lambda_j + \rho_j \text{Var}(v|i_j, p_j)} \quad (55)$$

$$\stackrel{\text{FOC}}{\implies} (2\lambda_j + \rho_j \text{Var}(v|i_j, p_j))p = (\lambda_j + \rho_j \text{Var}(v|i_j, p_j))p_j + \lambda_j \mathbb{E}(v|i_j, p_j) \quad (56)$$

$$\Rightarrow p_j = \frac{(2\lambda_j + \rho_j \text{Var}(v|i_j, p_j))p - \lambda_j \mathbb{E}(v|i_j, p_j)}{(\lambda_j + \rho_j \text{Var}(v|i_j, p_j))} \quad (57)$$

thus (given that p_j and p are informationally equivalent)

$$x_j = \frac{\mathbb{E}(v|i_j, p) - p}{(\lambda_j + \rho_j \text{Var}(v|i_j, p))}$$

Conjecture a linear strategy profile $x_j = \mu_j + \beta_j i_j - \gamma_j p$, then by market clearing

$$\sum_{k=1}^N \mu_k + \sum_{k=1}^N \beta_k i_k - \sum_{k=1}^N \gamma_k p + z = 0 \quad (58)$$

or

$$x_j + \sum_{k \neq j} \mu_k + \sum_{k \neq j} \beta_k i_k - \sum_{k \neq j} \gamma_k p + z = 0 \quad (59)$$

$$\Rightarrow p = \frac{x_j + \sum_{k \neq j} \mu_k + \sum_{k \neq j} \beta_k i_k + z}{\sum_{k \neq j} \gamma_k} \quad (60)$$

$$\Rightarrow \lambda_j = \frac{1}{\sum_{k \neq j} \gamma_k}, p_j = \frac{\sum_{k \neq j} \mu_k + \sum_{k \neq j} \beta_k i_k + z}{\sum_{k \neq j} \gamma_k} \quad (61)$$

thus

$$\mathbb{E}(v|i_j, p_j) = \bar{v} + \frac{(\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2)(i_j - \bar{v}) + \sum_{k \neq j} \beta_k \tau_j^{-1} [\sum_{k \neq j} \beta_k (i_k - \bar{v}) + z - \bar{z}]}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2} \quad (62)$$

$$\text{Var}(v|i_j, p_j) = \frac{\tau_j^{-1} [\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2]}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2} \quad (63)$$

and

$$x_j = \frac{\bar{v} + \frac{(\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2)(i_j - \bar{v}) + \sum_{k \neq j} \beta_k \tau_j^{-1} [\sum_{k \neq j} \beta_k (i_k - \bar{v}) + z - \bar{z}]}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2} - \frac{\sum_{k \neq j} \mu_k + \sum_{k \neq j} \beta_k i_k + z}{\sum_{k \neq j} \gamma_k}}{2 \frac{1}{\sum_{k \neq j} \gamma_k} + \rho_j \frac{\tau_j^{-1} [\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2]}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2}} \quad (64)$$

$$= \mu_j + \beta_j i_j - \gamma_j p \quad (65)$$

$$= \mu_j + \beta_j i_j - \gamma_j \frac{\sum_{k=1}^N \mu_k + \sum_{k=1}^N \beta_k i_k + z}{\sum_{k=1}^N \gamma_k} \quad (66)$$

Equating coefficients we get

$$\bar{v} + \frac{(\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2)(-\bar{v}) + \sum_{k \neq j} \beta_k \tau_j^{-1} [\sum_{k \neq j} \beta_k (-\bar{v}) - \bar{z}]}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2} - \frac{\sum_{k \neq j} \mu_k}{\sum_{k \neq j} \gamma_k} = \mu_j - \gamma_j \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k} \quad (67)$$

$$2 \frac{1}{\sum_{k \neq j} \gamma_k} + \rho_j \frac{\tau_j^{-1} [\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2]}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2} = \beta_j - \gamma_j \frac{\beta_j}{\sum_{k=1}^N \gamma_k} \quad (68)$$

$$2 \frac{1}{\sum_{k \neq j} \gamma_k} + \rho_j \frac{\tau_j^{-1} [\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2]}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2} = - \frac{\gamma_j}{\sum_{k=1}^N \gamma_k} \quad (69)$$

Equation (68) and (69) lead to

$$\frac{[\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2](\sum_{k=1}^N \gamma_k - \sum_{k \neq j} \gamma_k \beta_j \rho_j \tau_j^{-1})}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2} = 2\beta_j \quad (70)$$

$$\frac{\sum_{k=1}^N \gamma_k \sum_{k \neq j} \beta_k \tau_j^{-1} + \gamma_j \rho_j \tau_j^{-1} [\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2]}{\tau_j^{-1} [(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2} = \frac{-\gamma_j}{\sum_{k \neq j} \gamma_k} + 1 \quad (71)$$

(67) leads to

$$\begin{aligned} & \tau_v \tau_j^{-1} \left[\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2 \right] \bar{v} - \sum_{k \neq j} \beta_k \tau_j^{-1} \bar{z} = (\mu_j - \gamma_j \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k}) \rho_j \tau_j^{-1} \left[\sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2 \right] \\ & + \left[(\mu_j - \gamma_j \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k}) \frac{1}{\sum_{k \neq j} \gamma_k} + \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k} \right] \left[\tau_j^{-1} \left[(\sum_{k \neq j} \beta_k)^2 + \tau_v \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2 \right] + \sum_{k \neq j} \beta_k^2 \tau_k^{-1} + \sigma_z^2 \right] \end{aligned} \quad (72)$$

The three equations above defines the equilibrium.

Consider a completely inefficient market ($\sigma_z^2 = +\infty$), we have

$$\begin{aligned}\frac{(\tau_j \sum_{k=1}^N \gamma_k - \sum_{k \neq j} \gamma_k \beta_j \rho_j)}{\tau_j + \tau_v} &= 2\beta_j \\ \frac{\gamma_j \rho_j}{\tau_j + \tau_v} &= \frac{-\gamma_j}{\sum_{k \neq j} \gamma_k} + 1 \\ \tau_v \bar{v} &= (\mu_j - \gamma_j \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k}) \rho_j + [(\mu_j - \gamma_j \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k}) \frac{1}{\sum_{k \neq j} \gamma_k} + \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k}] (\tau_j + \tau_v)\end{aligned}$$

By $\gamma_j \leq \sum_{k=1}^N \gamma_k$ (deleting one positive root), we have

$$\gamma_j = \frac{C}{2} + \frac{\tau_j + \tau_v}{\rho_j} - \sqrt{\frac{C^2}{4} + (\frac{\tau_j + \tau_v}{\rho_j})^2}, \text{ where } C = \sum_{k=1}^N \gamma_k \quad (73)$$

and

$$\beta_j = \frac{\tau_j}{\tau_j + \tau_v} \gamma_j \quad (74)$$

$$\mu_j = \frac{\tau_v}{\rho_j + \frac{1}{C - \gamma_j} (\tau_j + \tau_v)} \bar{v} \quad (75)$$

Or in a symmetric economy ($\rho_j = \rho, \tau_j = \tau_e, \gamma_j = \gamma, \beta_j = \beta$)

$$\gamma = \frac{\tau_e^{-1} \tau_v [(N-1)\beta^2 \tau_e^{-1} + \sigma_z^2]}{N(N-1)\beta^2 \tau_e^{-1} + \sigma_z^2} \beta + \beta \quad (76)$$

$$\beta^3 (N-1)^2 \rho + \beta^2 (N-1) N \tau_e + \beta (N-1) \rho \sigma_z^2 \tau_e - (N-2) \sigma_z^2 \tau_e^2 = 0 \quad (77)$$

(77) increases in β , thus $0 < \beta < \min\{\frac{N-2}{N-1} \frac{\tau_e}{\rho}, \frac{\sqrt{\tau_e \sigma_z}}{\sqrt{N}}\}$

Revenue-sharing Now consider $M \leq N$ investors agree to partner. Perceiving a residual supply curve $p = p_j + \lambda_j x_j$, partner j chooses x_j to maximize

$$\mathbb{E} \left[-\exp(-\rho_j a_j (v - p_j - \lambda_j x_j) (x_j + \sum_{k \neq j, k=1}^M x_k)) | i_j, p \right],$$

Conjecture a linear strategy profile $x_j = \mu_j + \beta_j i_j - \gamma_j p$, thus

$$x_j + \sum_{k \neq j, k=1}^M x_k = x_j + \sum_{k \neq j, k=1}^M \mu_k + \sum_{k \neq j, k=1}^M \beta_k i_k - \sum_{k \neq j, k=1}^M \gamma_k p \quad (78)$$

and

$$\left(\begin{array}{c} -\rho_j a_j (v - p_j - \lambda_j x_j) \\ x_j + \sum_{k \neq j, k=1}^M x_k \end{array} \right)_{|i_j, p} \sim \mathcal{N} \left(\begin{array}{c} \left[\begin{array}{c} -\rho_j a_j (\mathbb{E}(v|i_j, p) - p_j - \lambda_j x_j) \\ x_j + \sum_{k \neq j, k=1}^M \mu_k + \sum_{k \neq j, k=1}^M \beta_k \mathbb{E}(v|i_j, p) - \sum_{k \neq j, k=1}^M \gamma_k p \end{array} \right], \\ \left[\begin{array}{cc} \rho_j^2 a_j^2 \text{Var}(v|i_j, p) & -\rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) \\ -\rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) & (\sum_{k \neq j, k=1}^M \beta_k)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1} \end{array} \right] \end{array} \right) \quad (79)$$

whose certainty equivalent is (by Lemma A.1)

$$- \frac{\exp(\frac{A}{2B})}{\sqrt{B}}, \quad (80)$$

where

$$\begin{aligned} A &= \rho_j^2 a_j^2 (\mathbb{E}(v|i_j, p) - p_j - \lambda_j x_j)^2 \left[\left(\sum_{k \neq j, k=1}^M \beta_k \right)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1} \right] \\ &+ \left[x_j + \sum_{k \neq j, k=1}^M \mu_k + \sum_{k \neq j, k=1}^M \beta_k \mathbb{E}(v|i_j, p) - \sum_{k \neq j, k=1}^M \gamma_k p \right]^2 \rho_j^2 a_j^2 \text{Var}(v|i_j, p) \\ &- 2\rho_j a_j (\mathbb{E}(v|i_j, p) - p_j - \lambda_j x_j) \left[x_j + \sum_{k \neq j, k=1}^M \mu_k + \sum_{k \neq j, k=1}^M \beta_k \mathbb{E}(v|i_j, p) - \sum_{k \neq j, k=1}^M \gamma_k p \right] \\ &\quad \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) \right] \\ B &= \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) \right]^2 \\ &- \rho_j^2 a_j^2 \text{Var}(v|i_j, p) \left[\left(\sum_{k \neq j, k=1}^M \beta_k \right)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1} \right] \end{aligned}$$

Taking FOC w.r.t. x_j we get

$$\begin{aligned} &- \lambda_j \rho_j^2 a_j^2 (\mathbb{E}(v|i_j, p) - p_j - \lambda_j x_j) \left[\left(\sum_{k \neq j, k=1}^M \beta_k \right)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1} \right] \\ &+ \left[x_j + \sum_{k \neq j, k=1}^M \mu_k + \sum_{k \neq j, k=1}^M \beta_k \mathbb{E}(v|i_j, p) - \sum_{k \neq j, k=1}^M \gamma_k p \right] \rho_j^2 a_j^2 \text{Var}(v|i_j, p) \\ &+ \rho_j a_j \lambda_j \left[x_j + \sum_{k \neq j, k=1}^M \mu_k + \sum_{k \neq j, k=1}^M \beta_k \mathbb{E}(v|i_j, p) - \sum_{k \neq j, k=1}^M \gamma_k p \right] \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) \right] \\ &- \rho_j a_j (\mathbb{E}(v|i_j, p) - p_j - \lambda_j x_j) \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) \right] = 0. \end{aligned}$$

thus plug in $x_j = \mu_j + \beta_j i_j - \gamma_j p$, and since

$$\begin{aligned}
p &= \frac{x_j + \sum_{k \neq j, k=1}^N \mu_k + \sum_{k \neq j, k=1}^N \beta_k i_k + z}{\sum_{k \neq j, k=1}^N \gamma_k} \\
&= \frac{\sum_{k=1}^N \mu_k + \sum_{k=1}^N \beta_k i_k + z}{\sum_{k=1}^N \gamma_k} \\
\lambda_j &= \frac{1}{\sum_{k \neq j, k=1}^N \gamma_k} \\
p_j &= \frac{\sum_{k \neq j, k=1}^N \mu_k + \sum_{k \neq j, k=1}^N \beta_k i_k + z}{\sum_{k \neq j, k=1}^N \gamma_k} \\
\mathbb{E}(v|i_j, p_j) &= \bar{v} + \frac{(\sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2)(i_j - \bar{v}) + \sum_{k \neq j, k=1}^N \beta_k \tau_j^{-1} [\sum_{k \neq j} \beta_k (i_k - \bar{v}) + z - \bar{z}]}{\tau_j^{-1} [(\sum_{k \neq j, k=1}^N \beta_k)^2 + \tau_v \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2} \\
\text{Var}(v|i_j, p_j) &= \frac{\tau_j^{-1} [\sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2]}{\tau_j^{-1} [(\sum_{k \neq j, k=1}^N \beta_k)^2 + \tau_v \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2}
\end{aligned}$$

Equating coefficients we get

Constants:

$$\begin{aligned}
& \left[\sum_{k=1}^M \mu_k + \left(\sum_{k \neq j, k=1}^M \beta_k \right) \bar{v} - \left(\sum_{k \neq j, k=1}^M \beta_k \right) \frac{(\sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2) \bar{v} + \sum_{k \neq j, k=1}^N \beta_k \tau_j^{-1} [\sum_{k \neq j} \beta_k \bar{v} + \bar{z}]}{\tau_j^{-1} [(\sum_{k \neq j, k=1}^N \beta_k)^2 + \tau_v \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2} \right. \\
& \quad \left. - \sum_{k=1}^M \gamma_k \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k} \right] \left[\rho_j a_j \text{Var}(v|i_j, p) + \lambda_j + \lambda_j \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) \right] \\
= & \left(\bar{v} + \frac{(\sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2)(-\bar{v}) + \sum_{k \neq j, k=1}^N \beta_k \tau_j^{-1} [\sum_{k \neq j} \beta_k (-\bar{v}) - \bar{z}]}{\tau_j^{-1} [(\sum_{k \neq j, k=1}^N \beta_k)^2 + \tau_v \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2} - \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k} \right) \\
& \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) + \lambda_j \rho_j a_j [(\sum_{k \neq j, k=1}^M \beta_k)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1}] \right] \\
i_j : & \\
& \left[\beta_j + \left(\sum_{k \neq j, k=1}^M \beta_k \right) \frac{(\sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2)}{\tau_j^{-1} [(\sum_{k \neq j, k=1}^N \beta_k)^2 + \tau_v \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2} - \sum_{k=1}^M \gamma_k \frac{\beta_j}{\sum_{k=1}^N \gamma_k} \right] \\
& \left[\rho_j a_j \text{Var}(v|i_j, p) + \lambda_j + \lambda_j \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) \right] \\
= & \left(\frac{(\sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2)}{\tau_j^{-1} [(\sum_{k \neq j, k=1}^N \beta_k)^2 + \tau_v \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2} - \frac{\beta_j}{\sum_{k=1}^N \gamma_k} \right) \\
& \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) + \lambda_j \rho_j a_j [(\sum_{k \neq j, k=1}^M \beta_k)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1}] \right] \\
z : & \\
& \left[\left(\sum_{k \neq j, k=1}^M \beta_k \right) \frac{\sum_{k \neq j, k=1}^N \beta_k \tau_j^{-1}}{\tau_j^{-1} [(\sum_{k \neq j, k=1}^N \beta_k)^2 + \tau_v \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2} - \frac{\sum_{k=1}^M \gamma_k}{\sum_{k=1}^N \gamma_k} \right] \\
& \left[\rho_j a_j \text{Var}(v|i_j, p) + \lambda_j + \lambda_j \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) \right] \\
= & \left(\frac{\sum_{k \neq j, k=1}^N \beta_k \tau_j^{-1}}{\tau_j^{-1} [(\sum_{k \neq j, k=1}^N \beta_k)^2 + \tau_v \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \tau_v \sigma_z^2] + \sum_{k \neq j, k=1}^N \beta_k^2 \tau_k^{-1} + \sigma_z^2} - \frac{1}{\sum_{k=1}^N \gamma_k} \right) \\
& \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \text{Var}(v|i_j, p) + \lambda_j \rho_j a_j [(\sum_{k \neq j, k=1}^M \beta_k)^2 \text{Var}(v|i_j, p) + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1}] \right]
\end{aligned}$$

Special case 1: $\sigma_z^2 \rightarrow \infty$ Equating coefficients we get $\forall 1 \leq j \leq M$

$$\begin{aligned}
& \left[\sum_{k=1}^M \mu_k + \left(\sum_{k \neq j, k=1}^M \beta_k \right) \frac{\tau_v}{\tau_v + \tau_j} \bar{v} - \sum_{k=1}^M \gamma_k \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k} \right] \left[\rho_j a_j \frac{1}{\tau_v + \tau_j} + \lambda_j + \lambda_j \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \frac{1}{\tau_v + \tau_j} \right] \\
&= \left(\frac{\tau_v}{\tau_v + \tau_j} \bar{v} - \frac{\sum_{k=1}^N \mu_k}{\sum_{k=1}^N \gamma_k} \right) \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \frac{1}{\tau_v + \tau_j} + \lambda_j \rho_j a_j \left[\left(\sum_{k \neq j, k=1}^M \beta_k \right)^2 \frac{1}{\tau_v + \tau_j} + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1} \right] \right] \\
& \quad \left[\beta_j + \left(\sum_{k \neq j, k=1}^M \beta_k \right) \frac{\tau_j}{\tau_v + \tau_j} - \sum_{k=1}^M \gamma_k \frac{\beta_j}{\sum_{k=1}^N \gamma_k} \right] \left[\rho_j a_j \frac{1}{\tau_v + \tau_j} + \lambda_j + \lambda_j \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \frac{1}{\tau_v + \tau_j} \right] \\
&= \left(\frac{\tau_j}{\tau_v + \tau_j} - \frac{\beta_j}{\sum_{k=1}^N \gamma_k} \right) \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \frac{1}{\tau_v + \tau_j} + \lambda_j \rho_j a_j \left[\left(\sum_{k \neq j, k=1}^M \beta_k \right)^2 \frac{1}{\tau_v + \tau_j} + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1} \right] \right] \\
& \quad \sum_{k=1}^M \gamma_k \left[\rho_j a_j \frac{1}{\tau_v + \tau_j} + \lambda_j + \lambda_j \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \frac{1}{\tau_v + \tau_j} \right] \\
&= \left[1 + \rho_j a_j \sum_{k \neq j, k=1}^M \beta_k \frac{1}{\tau_v + \tau_j} + \lambda_j \rho_j a_j \left[\left(\sum_{k \neq j, k=1}^M \beta_k \right)^2 \frac{1}{\tau_v + \tau_j} + \sum_{k \neq j, k=1}^M \beta_k^2 \tau_k^{-1} \right] \right] \\
\lambda_j &= \frac{1}{\sum_{k \neq j, k=1}^N \gamma_k}
\end{aligned}$$

and $\forall M+1 \leq j \leq N$

$$\gamma_j = \frac{C}{2} + \frac{\tau_j + \tau_v}{\rho_j} - \sqrt{\frac{C^2}{4} + \left(\frac{\tau_j + \tau_v}{\rho_j} \right)^2}, \text{ where } C = \sum_{k=1}^N \gamma_k \quad (81)$$

$$\beta_j = \frac{\tau_j}{\tau_j + \tau_v} \gamma_j \quad (82)$$

$$\mu_j = \frac{\tau_v}{\rho_j + \frac{1}{C - \gamma_j} (\tau_j + \tau_v)} \bar{v} \quad (83)$$

A special case: *ex ante* identical investors $\forall 1 \leq j \leq M$

$$\begin{aligned}
& [M\mu_1 + (M-1)\beta_1 \frac{\tau_v}{\tau_v + \tau_e} \bar{v} - M\gamma_1 \frac{M\mu_1 + (N-M)\mu_2}{M\gamma_1 + (N-M)\gamma_2}] \left[\rho \frac{1}{M} \frac{1}{\tau_v + \tau_e} + \lambda_1 + \lambda_1 \rho \frac{1}{M} \frac{(M-1)\beta_1}{\tau_v + \tau_e} \right] \\
= & \left(\frac{\tau_v}{\tau_v + \tau_e} \bar{v} - \frac{M\mu_1 + (N-M)\mu_2}{M\gamma_1 + (N-M)\gamma_2} \right) \left[1 + \rho \frac{1}{M} \frac{(M-1)\beta_1}{\tau_v + \tau_e} + \lambda_1 \rho \frac{1}{M} [((M-1)\beta_1)^2 \frac{1}{\tau_v + \tau_e} + (M-1)\beta_1^2 \tau_e^{-1}] \right] \\
& [\beta_1 + (M-1)\beta_1 \frac{\tau_e}{\tau_v + \tau_e} - M\gamma_1 \frac{\beta_1}{M\gamma_1 + (N-M)\gamma_2}] \left[\rho \frac{1}{M} \frac{1}{\tau_v + \tau_e} + \lambda_1 + \lambda_1 \rho \frac{1}{M} \frac{(M-1)\beta_1}{\tau_v + \tau_e} \right] \\
= & \left(\frac{\tau_e}{\tau_v + \tau_e} - \frac{\beta_1}{M\gamma_1 + (N-M)\gamma_2} \right) \left[1 + \rho \frac{1}{M} \frac{(M-1)\beta_1}{\tau_v + \tau_e} + \lambda_1 \rho \frac{1}{M} [((M-1)\beta_1)^2 \frac{1}{\tau_v + \tau_e} + (M-1)\beta_1^2 \tau_e^{-1}] \right] \\
& M\gamma_1 \left[\rho \frac{1}{M} \frac{1}{\tau_v + \tau_e} + \lambda_1 + \lambda_1 \rho \frac{1}{M} \frac{(M-1)\beta_1}{\tau_v + \tau_e} \right] \\
= & \left[1 + \rho \frac{1}{M} \frac{(M-1)\beta_1}{\tau_v + \tau_e} + \lambda_1 \rho \frac{1}{M} [((M-1)\beta_1)^2 \frac{1}{\tau_v + \tau_e} + (M-1)\beta_1^2 \tau_e^{-1}] \right] \\
\lambda_1 = & \frac{1}{(M-1)\gamma_1 + (N-M)\gamma_2}
\end{aligned}$$

and $\forall M+1 \leq j \leq N$

$$\gamma_2 = \frac{C}{2} + \frac{\tau_e + \tau_v}{\rho} - \sqrt{\frac{C^2}{4} + \left(\frac{\tau_e + \tau_v}{\rho}\right)^2}, \text{ where } C = M\gamma_1 + (N-M)\gamma_2 \quad (84)$$

$$\beta_2 = \frac{\tau_e}{\tau_e + \tau_v} \gamma_2 \quad (85)$$

$$\mu_2 = \frac{\tau_v}{\rho + \frac{1}{C-\gamma_2}(\tau_e + \tau_v)} \bar{v} \quad (86)$$

$\forall 1 \leq j \leq M$

$$\beta_1 = \frac{M\tau_e}{\tau_v + M\tau_e} \gamma_1$$

$$\gamma_1 = \frac{C}{2} + \frac{\tau_v + M\tau_e}{2\rho} - \sqrt{\frac{C^2}{4} + \left(\frac{M\tau_e + \tau_v}{\rho}\right)^2}$$

(deleting one root)

$$C = M \left(\frac{C}{2} + \frac{\tau_v + M\tau_e}{2\rho} - \sqrt{\frac{C^2}{4} + \left(\frac{M\tau_e + \tau_v}{\rho}\right)^2} \right) + (N-M) \left(\frac{C}{2} + \frac{\tau_e + \tau_v}{\rho} - \sqrt{\frac{C^2}{4} + \left(\frac{\tau_e + \tau_v}{\rho}\right)^2} \right)$$

Full information: M out of N homogeneous investors share information From (70), (71), and (72), $\forall 1 \leq j \leq M$

$$\frac{\left[\frac{M-1}{M}\beta_1^2 + (N-M)\beta_2^2 + \tau_e\sigma_z^2\right] [C\tau_e - (C - \gamma_1)\beta_1\rho/M]}{\tau_e[(M-1)\beta_1 + (N-M)\beta_2]^2 + (\tau_v + M\tau_e) \left[\frac{M-1}{M}\beta_1^2 + (N-M)\beta_2^2 + \tau_e\sigma_z^2\right]} = 2\beta_1/M$$

$$\frac{C\tau_e[(M-1)\beta_1 + (N-M)\beta_2] + \gamma_1\rho\left[\frac{M-1}{M}\beta_1^2 + (N-M)\beta_2^2 + \tau_e\sigma_z^2\right]}{\tau_e[(M-1)\beta_1 + (N-M)\beta_2]^2 + (\tau_v + M\tau_e) \left[\frac{M-1}{M}\beta_1^2 + (N-M)\beta_2^2 + \tau_e\sigma_z^2\right]} = \frac{-\gamma_1}{C - \gamma_1} + 1$$

$\forall M+1 \leq j \leq N$

$$\frac{[\beta_1^2 + (N-M-1)\beta_2^2 + \tau_e\sigma_z^2][C\tau_e - (C - \gamma_2)\beta_2\rho]}{\tau_e[M\beta_1 + (N-M-1)\beta_2]^2 + (\tau_v + \tau_e)[\beta_1^2 + (N-M-1)\beta_2^2 + \tau_e\sigma_z^2]} = 2\beta_2$$

$$\frac{C\tau_e[M\beta_1 + (N-M-1)\beta_2] + \gamma_2\rho[\beta_1^2 + (N-M-1)\beta_2^2 + \tau_e\sigma_z^2]}{\tau_e[M\beta_1 + (N-M-1)\beta_2]^2 + (\tau_v + \tau_e)[\beta_1^2 + (N-M-1)\beta_2^2 + \tau_e\sigma_z^2]} = \frac{-\gamma_2}{C - \gamma_2} + 1$$

As $\sigma_z^2 \rightarrow +\infty$, we have $\forall 1 \leq j \leq M$

$$\beta_1 = \frac{M\tau_e}{\tau_v + M\tau_e}\gamma_1$$

$$\gamma_1 = \frac{C}{2} + \frac{(\tau_v + M\tau_e)}{\rho} - \sqrt{\frac{C^2}{4} + \left(\frac{\tau_e + M\tau_v}{\rho}\right)^2}$$

$\forall M+1 \leq j \leq N$

$$\beta_2 = \frac{\tau_e}{\tau_e + \tau_v}\gamma_2$$

$$\gamma_2 = \frac{C}{2} + \frac{\tau_e + \tau_v}{\rho} - \sqrt{\frac{C^2}{4} + \left(\frac{\tau_e + \tau_v}{\rho}\right)^2}, \text{ where } C = M\gamma_1 + (N-M)\gamma_2$$

thus payoff is $\mathbb{E}[-\exp(-\rho_j(v - p(x_j))x_j)|i_j, p]$

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